DynSphere: State and Motion Consistency for Multivariate Telemetry Anomaly Detection

Supplementary Evaluations

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1 Scope

This supplement reports additional studies and details that support the main paper. It includes a 13-method study under the TSB-AD-M protocol, held-out diagnostics for DynSphere and the statistical baselines, the PowerSystem diagnostic, supporting accuracy results, a PCA-dimension sensitivity study, CPU quality and cost measurements, and the full controlled-explanation and representation study. Each study keeps its own fitting, scoring, and aggregation protocol, and all tables are generated directly from the stored results. The main paper specifies the method and the primary evaluation.

We use the configuration names of the main paper: S1 and S4 are the spherical configurations with one and four concepts, and E1 and E4 are the Euclidean configurations. Tables label them DynSphere-S1, DynSphere-S4, DynSphere-E1, and DynSphere-E4.

2 TSB-AD-M study: setup

2.1 Datasets and training protocols

This study follows the TSB-AD-M protocol, in which the scored sequence includes the training observations. It uses six real datasets and the DynSphere-Contextual benchmark as a diagnostic. Table S1 reports the processed array sizes. MSL and SMAP contain spacecraft telemetry [3], and SMD contains server metrics [9]. We evaluate every MSL, SMAP, and SMD file in the TSB-AD-M evaluation list [4], i.e., 14, 25, and 20 series, respectively. These 59 files are subsets of the original releases.

For these files, training uses the training-segment length encoded in the filename, and the full series, including the training segment, is scored and evaluated. Training follows the benchmark specification, without removing labeled anomalies. A label check found three anomalous rows in the 500-row training segment of SMD file 078; we keep this file and report the issue. This protocol therefore measures ranking on the benchmark with a small, known contamination of the training data, and the scored sequence includes the training observations.

BATADAL contains measurements from a water distribution network [10]. We train on the normal rows of dataset03 and score dataset04. HAI-20.07 contains measurements from coupled industrial processes [8]. We train on all normal rows of train1 and score all rows of test1, using the full processed HAI arrays reported in Table S1.

PowerSystem contains industrial power-system scenarios [2, 5]. The normal-labeled rows of the combined binary source form its training array, and the complete combined source is scored. Its normal training rows therefore also occur in the evaluation sequence, so this protocol evaluates against an in-sample normal reference. Its high anomaly prevalence (71.02%) also affects the interpretation of its precision-recall values.

DynSphere-Contextual records the mechanism that generates each anomalous row. This information is unavailable for the real datasets, so we use the benchmark to diagnose which types of anomaly each method detects. Its results are excluded from the real-dataset means. The benchmark includes 24 training entities and 12 test entities. Each entity follows a hidden schedule of small libration, wide libration, precessing motion, and near-resonant motion. The observed variables are twelve noisy nonlinear functions of latent position, velocity, acceleration, and energy. The labeled anomalies are generated by four latent edits, namely teleportation, phase lag, velocity inversion, and cross-concept copying, which affect 294, 415, 230, and 216 rows, respectively. The feature arrays contain only the observed `x_` columns. Latent states, mechanism identifiers, and labels are excluded from model inputs.

The controlled benchmark tests deviations from its prescribed trajectory structure. It does not guarantee that anomalous and normal rows have identical marginal distributions. This study sorts entities by identifier and time and passes their concatenated arrays to every method as a single sequence. Derivative and window computations can therefore span the boundaries between entities. The example in Figure 1 of the main paper comes from a separate analysis that keeps entity identifiers. It illustrates spherical state and motion evidence under that sequence handling, and its trace is not part of the scores of this study.

2.2 Methods and configurations

The statistical baselines use the same definitions in every study: PCA keeps at least 95% of the fitting variance and scores squared reconstruction residuals; k-means uses four centers and scores the distance to the nearest center. The primary study adds a ridge AR(1) predictor with penalty 1, fitted to successive observations within each entity and scored by the mean squared next-step error. This baseline is distinct from the exploratory neural forecasting baseline.

Table S1: Data used in the TSB-AD-M study. For MSL, SMAP, and SMD, training and scored rows are summed over the selected series. Contextual denotes DynSphere-Contextual. For MSL, SMAP, SMD, and PowerSystem, training rows overlap with scored rows.

Dataset	Series	Train rows	Scored rows	Features	Anomalous rows	Training selection
BATADAL	1	8,761	4,177	43	219	Normal rows of dataset03
HAI-20.07	1	309,600	291,600	59	11,538	Normal rows of train1
PowerSystem	1	22,714	78,377	128	55,663	Normal rows of combined source
MSL	14	10,497	41,380	55	2,098	Training segment
SMAP	25	48,771	195,690	25	5,790	Training segment
SMD	20	66,589	503,059	38	16,699	Training segment
Contextual	1	57,600	28,800	12	1,155	Released normal training file

The TSB-AD-M study includes PCA reconstruction, k-means distance, DAGMM, CATCH, MTAD-GAT, OmniAnomaly, Anomaly Transformer, TranAD, the one-step Euclidean neural forecasting baseline, and the four DynSphere configurations.

External methods use fixed code versions. DAGMM uses the implementation available in TAB [7, 15]. CATCH, Anomaly Transformer, and TranAD use the repositories released by their authors [11–13]. OmniAnomaly uses the TSB-AD implementation [4, 9]. MTAD-GAT uses the ML4ITS PyTorch implementation [14], whose GATv2 layers and gated recurrent unit (GRU) reconstruction decoder differ from the architecture described in the original paper. For this reason, we refer to it as MTAD-GAT (ML4ITS) in every table.

A common wrapper gives every method the same training and scoring procedure, feature order, and score direction. It passes standardized arrays and no evaluation labels. For CATCH, we correct the averaging of frequency patches: the average at each position divides by the number of windows that cover it, including windows with zero loss, which prevents division by zero. Each run records the code version of its method and of the wrapper (`1160ee7`). Our repository records the complete commit identifiers, dependency versions, and configurations.

Training budgets are ten epochs for DAGMM, CATCH, and MTAD-GAT, 20 for OmniAnomaly, three for Anomaly Transformer, and five for TranAD. The one-step forecasting baseline uses 15 epochs, one mode, latent dimension $\min(6, d)$, and hidden dimension 48. The results describe these implementations under the stated budgets and do not attempt to reproduce the tuned results of the original papers. In particular, a low score for a competitor reflects its configuration and these datasets.

DynSphere uses $q = \min(5, d)$, a ridge penalty of 1, a transition smoothing of 0.5, and a propagation radius of 8. The weights for position, latent residual, raw residual, joint state and speed, speed, transition, and cluster distance are 1, 1.8, 1.4, 1, 0.8, 0.8, and 0.15, respectively. These weights are shared by all four DynSphere configurations. We report every configuration, including those that perform less favorably.

The one-step Euclidean forecasting baseline is an exploratory diagnostic maintained in our repository. It learns a probabilistic next-state model with one mode and scores predictive inconsistency. Its training objective and scoring procedure differ from the local acceleration models used by DynSphere. The primary held-out comparison uses linear ridge prediction as its one-step temporal baseline.

2.3 Metrics, coverage, and aggregation

Average precision (AP), reported as AUPRC in the tables, full AUROC, the volume under the precision-recall surface (VUS-PR), and the volume under the ROC surface (VUS-ROC) provide secondary evaluations. All 2,453 valid runs are evaluated with the official VUS implementation [1, 6]. Its maximum buffer width is set to the median length of the labeled events. This label-derived width is used only by the evaluator. No method receives labels, and we do not use point adjustment. Metric values are averaged over seeds within each series and then over series within each dataset. The tables report these dataset-level means. Means over datasets give the six real datasets equal weight. For completeness, Table S2 also reports AUPRC and VUS-PR means over all seven datasets, i.e., including DynSphere-Contextual. Since the AUPRC baseline depends on anomaly prevalence, we accompany each mean with dataset-level results.

The study attempts $63 \times 3 \times 13 = 2{,}457$ unique runs and obtains 2,453 valid score vectors. DAGMM produces non-finite scores for SMAP file 159 at all three seeds and for file 169 at seed 38, and identical retries reproduce these four failures. Its coverage is 185/189 overall and 71/75 on SMAP. All other methods complete 189/189 runs. Failed runs are recorded without imputing values.

The comparison of all methods uses the 185 series-seed pairs that every method completes (2,405 method evaluations). After averaging over seeds, these pairs cover 62 series, 61 of which belong to the real datasets. The geometry analysis uses all series (62 real series and the contextual benchmark), since every DynSphere configuration completed every run. We also report means over each method's available runs to show the effect of restricting the comparison to the series completed by all methods.

Paired differences use the same equal-dataset mean, computed over the six real datasets and, for completeness, over all seven. We first average seeds within each series and then resample series with replacement within each dataset, using 20,000 bootstrap replicates and seed 20261019. This procedure preserves the pairing of model differences and treats each series as the sampling unit. The reported intervals are descriptive 95% percentile intervals, without correction for multiple comparisons, and they are conditional on the selected datasets. Datasets with a single series contribute no resampling

variation. The intervals therefore do not estimate uncertainty over new industrial deployments or new synthetic generations.

The experiments ran on the American University computing cluster. The four PyTorch competitors use one H100 PCIe GPU per run, whereas DAGMM, OmniAnomaly, the one-step forecasting baseline, DynSphere, and the statistical baselines run on CPUs. Our repository contains the list of inputs, the metrics of every run, and the coverage check. Tables are generated from the stored metrics, without retraining any model.

3 TSB-AD-M study: results

3.1 Overall comparison

Table S2 reports results on the series-seed pairs completed by all methods. On the six real datasets, E1 obtains the highest AUPRC (0.4429) and VUS-PR (0.4804). CATCH has the highest AUROC and VUS-ROC (0.7482 and 0.8353, respectively), and E4 has the highest standardized partial AUROC (0.6538). The ranking therefore depends on the evaluation criterion. E1 also obtains the highest AUPRC and VUS-PR when the contextual benchmark is included (0.3960 and 0.4352).

Table S3 compares means over each method's available runs with means over the pairs completed by all methods, on the real datasets. The AUPRC ordering does not change. Over available runs, E1 obtains an AUPRC of 0.4463 and a VUS-PR of 0.4838, followed in AUPRC by E4 and S4. The four DAGMM failures therefore have a small effect on the means of the other methods. We report both views for completeness.

On the real datasets, the AUPRC difference between E1 and CATCH is +0.0390, with descriptive bootstrap interval from 0.0150 to 0.0630 (Table S5). The corresponding VUS-PR interval includes zero. Relative to the one-step forecasting baseline, the differences are +0.0487 in AUPRC and +0.0422 in VUS-PR, and both intervals are above zero. These comparisons favor the proposed method under the evaluated configurations, but the evidence depends on the metric and on the conditional uncertainty model.

3.2 Results by dataset

Table S4 shows that no configuration leads on every dataset. On BATADAL, E1 has the highest AUPRC and E4 has the highest VUS-PR. On SMD, E4 leads on both primary metrics. These results support the use of the proposed state and motion scores for water-network and server telemetry under this protocol.

The other datasets favor different models. OmniAnomaly has the highest AUPRC and VUS-PR on HAI-20.07. On MSL, PCA reconstruction leads on both metrics, and the one-step forecasting baseline has the highest AUROC and VUS-ROC. On SMAP, CATCH leads on AUPRC and VUS-PR, and S4 obtains a higher AUPRC than the Euclidean configurations. On PowerSystem, MTAD-GAT leads on both primary metrics. Since training and scored normal rows overlap on this dataset, this result provides limited evidence about unseen normal operation.

3.3 Geometry and concept comparison

The paired comparisons in Table S5 separate the effect of geometry from that of concept count. On the real datasets, with four concepts, the Euclidean-minus-spherical differences are +0.0078 in AUPRC and +0.0078 in VUS-PR, and both intervals include zero. This study is therefore inconclusive about a general benefit of either geometry with four concepts.

With one concept, Euclidean geometry improves AUPRC by 0.0193 and VUS-PR by 0.0200, and both intervals are above zero. Four concepts do not improve the means under either geometry. Under spherical geometry, the four-minus-one differences are +0.0030 in AUPRC and +0.0046 in VUS-PR. Under Euclidean geometry, they are −0.0084 and −0.0077. All four intervals include zero. At the dataset level, S4 is stronger on SMAP and weaker or similar on the other datasets. When the contextual benchmark is included, the Euclidean four-minus-one differences become negative, with intervals below zero.

Under this protocol, a single Euclidean local model is a competitive configuration, and the study shows no consistent aggregate benefit from spherical geometry or multiple concepts. These choices therefore require justification for each application. The grouped analysis in Section 4.1 further examines which components contribute to E1's score and finds limited or negative gains from adding state evidence on several datasets. The ranking in this study alone cannot establish a general benefit from combining the terms.

3.4 Contextual-mechanism diagnostic

DynSphere-Contextual records the mechanism behind each anomalous row, which allows us to examine which types of anomaly each method detects. Overall, CATCH has the highest AUPRC on this benchmark (0.1227), and E1 has the highest VUS-PR (0.1638). Generic frequency reconstruction can also detect the prescribed temporal edits. The generator and the proposed method share a geometric motivation, so independent mechanism tests are important when interpreting this diagnostic.

For each method and mechanism, we compute AUROC and AUPRC on the normal rows together with the rows of that mechanism, and average the values over seeds. In the separate analysis that keeps entity identifiers (Section 2), S4 is strongest on cross-concept transitions (AUROC 0.828) and weakest on phase lag (AUROC 0.585), with latent teleportation (0.704) and velocity inversion (0.641) in between. Motion edits that leave the concept unchanged are therefore the most difficult mechanisms for the proposed method.

3.5 Component ablation and sensitivity study

A separate three-seed component study evaluated 16 MSL, 27 SMAP, and 22 SMD series, using training segments free of labeled anomalies. It used a larger selection of series and removed the anomalous rows from the contaminated SMD training segment. We report its results under its own protocol, as supplementary evidence about the score components. They are separate from the 59-file TSB-AD-M study with external competitors.

In that study, removing raw dynamics lowers AUPRC from 0.154 to 0.084 on MSL, from 0.236 to 0.090 on SMAP, and from 0.453 to 0.094 on SMD. Removing the transition penalty or the joint state and speed term produces smaller and mixed changes. These results identify raw acceleration residuals as a major source of evidence under that protocol. The extension and held-out analyses in the main paper show different effects for some terms, so this finding supports inspecting the components separately for each dataset.

Table S2: TSB-AD-M study: equal-dataset means on the series-seed pairs completed by all methods. Real columns average the six real datasets, and All columns also include DynSphere-Contextual. AP denotes AUPRC, and pAUC denotes the standardized partial AUROC for FPR≤5%. The highest value in each column is in bold. All methods are evaluated on the same 185 series-seed pairs.

Method	Real AP	Real VUS-PR	Real AUROC	Real VUS-ROC	Real pAUC	All AP	All VUS-PR
PCA reconstruction	0.3818	0.4273	0.6766	0.7699	0.6198	0.3340	0.3752
k-means distance	0.3822	0.4214	0.6805	0.7734	0.6187	0.3338	0.3691
DAGMM (TAB)	0.1942	0.2364	0.5701	0.6762	0.5208	0.1723	0.2104
CATCH	0.4039	0.4657	**0.7482**	**0.8353**	0.6362	0.3638	0.4225
MTAD-GAT (ML4ITS)	0.3887	0.4177	0.7356	0.8029	0.6239	0.3443	0.3737
OmniAnomaly (TSB-AD)	0.3756	0.4351	0.6807	0.7918	0.6197	0.3281	0.3808
Anomaly Transformer	0.1644	0.2003	0.5023	0.5488	0.5051	0.1466	0.1794
TranAD	0.1531	0.1947	0.5001	0.5453	0.4989	0.1371	0.1751
One-step predictor	0.3942	0.4382	0.7002	0.7905	0.6278	0.3499	0.3957
DynSphere-S4	0.4265	0.4648	0.7082	0.7831	0.6420	0.3807	0.4150
DynSphere-E4	0.4341	0.4723	0.7128	0.7837	**0.6538**	0.3839	0.4185
DynSphere-S1	0.4229	0.4597	0.7006	0.7804	0.6378	0.3786	0.4136
DynSphere-E1	**0.4429**	**0.4804**	0.7020	0.7783	0.6528	**0.3960**	**0.4352**

Table S3: TSB-AD-M study: means over the six real datasets and run coverage. Available-run values use every valid run of each method, and common values use the series-seed pairs completed by all methods. Valid runs count the 186 real-dataset runs of each method.

Method	Avail. AP	Avail. VUS-PR	Common AP	Common VUS-PR	Valid runs
PCA reconstruction	0.3814	0.4270	0.3818	0.4273	186/186
k-means distance	0.3818	0.4209	0.3822	0.4214	186/186
DAGMM (TAB)	0.1942	0.2364	0.1942	0.2364	182/186
CATCH	0.4059	0.4682	0.4039	0.4657	186/186
MTAD-GAT (ML4ITS)	0.3906	0.4204	0.3887	0.4177	186/186
OmniAnomaly (TSB-AD)	0.3770	0.4365	0.3756	0.4351	186/186
Anomaly Transformer	0.1646	0.2004	0.1644	0.2003	186/186
TranAD	0.1532	0.1948	0.1531	0.1947	186/186
One-step predictor	0.3958	0.4396	0.3942	0.4382	186/186
DynSphere-S4	0.4300	0.4683	0.4265	0.4648	186/186
DynSphere-E4	0.4378	0.4761	0.4341	0.4723	186/186
DynSphere-S1	0.4270	0.4637	0.4229	0.4597	186/186
DynSphere-E1	0.4463	0.4838	0.4429	0.4804	186/186

The same study varies the number of concepts, the ridge penalty, the propagation radius, and the score weights. The median rank correlation with the default configuration is 0.990 for AUROC and 0.994 for AUPRC. Propagation has the largest effect, since its removal lowers AUPRC by 0.042 on MSL and by 0.073 on SMAP. Noise and temporal-decimation tests produce dataset-dependent changes, including gains under small Gaussian perturbations. The evidence therefore supports a limited conclusion about configuration stability within that study, and naturally noisy or irregularly sampled telemetry requires further evaluation. The full ablation and stress-test records are available in our repository.

3.6 Recorded run time

Table S6 reports the elapsed time of the valid runs of the TSB-AD-M study, including data preparation and method execution. External methods also incur interpreter and framework startup costs. The four DynSphere configurations have median costs between 0.10 and 0.15 seconds per series, and total costs between 372.0 and 396.1 seconds over 189 runs. The one-step forecasting baseline has a median of 4.68 seconds, and the longest valid run is a CATCH run of approximately 9,007 seconds.

These measurements cover datasets with 12–128 features. Sequence lengths vary: `016_MSL` has 1,096 observations before boundary exclusions, and the HAI-20.07 test sequence has 291,600. Our repository also includes 72 scaling measurements, each run in a new process. For 50,000 fitting observations and 16 features, median E1 fitting time is 0.312 seconds, and scoring 20,000 observations takes 0.044 seconds. E4 takes 1.502 and 0.062 seconds, respectively. These measurements use one CPU thread and three repetitions per setting. Reported peak memory includes loaded libraries. They describe the proposed configurations and do not compare them with the deep methods on the same hardware.

Table S4: TSB-AD-M study: results by dataset on the series-seed pairs completed by all methods, reported as AUPRC/VUS-PR. Values are averaged over seeds within each series and then over series within each dataset. Bold identifies the highest unrounded value in each column for the corresponding metric. Contextual denotes DynSphere-Contextual.

Method	BATADAL	HAI	PowerSystem	MSL	SMAP	SMD	Contextual
PCA reconstruction	0.443/0.618	0.392/0.336	0.724/0.887	**0.291**/**0.306**	0.098/0.101	0.343/0.317	0.047/0.062
k-means distance	0.437/0.553	0.369/0.308	0.719/0.882	0.222/0.269	0.148/0.141	0.398/0.375	0.043/0.056
DAGMM (TAB)	0.091/0.124	0.039/0.053	0.715/0.886	0.129/0.159	0.127/0.126	0.064/0.071	0.041/0.054
CATCH	0.314/0.443	0.494/0.525	0.798/0.911	0.179/0.218	**0.273**/**0.296**	0.365/0.401	**0.123**/0.163
MTAD-GAT (ML4ITS)	0.412/0.446	0.368/0.324	**0.820**/**0.930**	0.171/0.217	0.209/0.239	0.353/0.351	0.078/0.110
OmniAnomaly (TSB-AD)	0.426/0.555	**0.510**/**0.550**	0.716/0.881	0.120/0.151	0.111/0.114	0.371/0.359	0.043/0.055
Anomaly Transformer	0.052/0.073	0.070/0.067	0.712/0.885	0.057/0.078	0.034/0.044	0.061/0.055	0.039/0.054
TranAD	0.053/0.074	0.040/0.055	0.710/0.885	0.053/0.075	0.028/0.041	0.034/0.039	0.041/0.057
One-step predictor	0.558/0.696	0.320/0.262	0.725/0.885	0.195/0.246	0.160/0.153	0.407/0.387	0.084/0.141
DynSphere-S4	0.606/0.767	0.360/0.265	0.802/0.918	0.136/0.176	0.231/0.235	0.424/0.428	0.106/0.117
DynSphere-E4	0.653/**0.796**	0.340/0.274	0.810/0.920	0.158/0.195	0.193/0.199	**0.450**/**0.450**	0.083/0.096
DynSphere-S1	0.640/0.759	0.379/0.307	0.792/0.915	0.141/0.182	0.164/0.172	0.423/0.424	0.113/0.137
DynSphere-E1	**0.663**/0.794	0.365/0.306	0.793/0.912	0.180/0.215	0.208/0.211	0.449/0.445	0.115/**0.164**

Table S5: TSB-AD-M study: paired differences of equal-dataset means with descriptive 95% bootstrap intervals over series. Real columns use the six real datasets, and All columns also include DynSphere-Contextual. The first four contrasts use all series (62 real series, 63 in All), and the last four use the series completed by all methods (61 real series, 62 in All). Seeds are averaged before resampling. Intervals are conditional on the fixed datasets and are not adjusted for multiple comparisons.

Contrast	Real ΔAP	Real ΔVUS-PR	All ΔAP	All ΔVUS-PR
E4-S4	+0.0078 [-0.0072, +0.0220]	+0.0078 [-0.0059, +0.0204]	+0.0035 [-0.0095, +0.0156]	+0.0037 [-0.0080, +0.0145]
E1-S1	+0.0193 [+0.0039, +0.0374]	+0.0200 [+0.0062, +0.0370]	+0.0168 [+0.0037, +0.0324]	+0.0210 [+0.0092, +0.0355]
S4-S1	+0.0030 [-0.0078, +0.0165]	+0.0046 [-0.0059, +0.0178]	+0.0016 [-0.0077, +0.0132]	+0.0010 [-0.0079, +0.0123]
E4-E1	-0.0084 [-0.0212, +0.0020]	-0.0077 [-0.0199, +0.0023]	-0.0118 [-0.0227, -0.0029]	-0.0163 [-0.0267, -0.0077]
E1-CATCH	+0.0390 [+0.0150, +0.0630]	+0.0147 [-0.0065, +0.0351]	+0.0323 [+0.0117, +0.0529]	+0.0127 [-0.0055, +0.0302]
E1-MTAD	+0.0542 [+0.0270, +0.0797]	+0.0627 [+0.0327, +0.0911]	+0.0517 [+0.0284, +0.0736]	+0.0615 [+0.0358, +0.0859]
E1-predictor	+0.0487 [+0.0233, +0.0717]	+0.0422 [+0.0151, +0.0664]	+0.0461 [+0.0244, +0.0659]	+0.0395 [+0.0163, +0.0603]
S4-predictor	+0.0323 [+0.0064, +0.0559]	+0.0265 [-0.0003, +0.0507]	+0.0308 [+0.0086, +0.0510]	+0.0193 [-0.0036, +0.0400]

Table S6: Recorded elapsed time for valid runs of the TSB-AD-M study. Each series and seed contributes one run. Sum adds the recorded times, including those of concurrent runs. Because devices differ and framework startup is included, these values do not allow a direct comparison of hardware efficiency. Failed DAGMM attempts are excluded.

Method	Device	Runs	Median (s)	Sum (s)
PCA reconstruction	CPU	189	0.04	32.6
k-means distance	CPU	189	0.05	42.9
DAGMM (TAB)	CPU	185	7.25	1688.2
CATCH	H100	189	25.77	40318.5
MTAD-GAT (ML4ITS)	H100	189	16.03	15579.9
OmniAnomaly (TSB-AD)	CPU	189	14.18	16618.8
Anomaly Transformer	H100	189	35.82	19437.8
TranAD	H100	189	6.88	2342.0
One-step predictor	CPU	189	4.68	3592.8
DynSphere-S4	CPU	189	0.12	372.0
DynSphere-E4	CPU	189	0.15	380.9
DynSphere-S1	CPU	189	0.10	380.7
DynSphere-E1	CPU	189	0.10	396.1

3.7 CPU quality and cost on the same hardware

This study runs all four DynSphere configurations, PCA reconstruction, KMeans distance, ridge AR(1), CATCH, and MTAD-GAT (ML4ITS) on the same CPU. Before fitting, we select from each of the five primary datasets the series with the median test length (the lower median for an even count), breaking ties by series identifier and without using anomaly scores. The selected series are BATADAL, HAI-20.07, `015_MSL`, `152_SMAP`, and `074_SMD`. Their fitting/test row counts are 7,008/4,177, 247,680/291,600, 460/1,728, 1,658/6,221, and 5,739/21,522, respectively. All seeds 37–39 use the full sequences and the original training budgets. The identifier of this study's run record begins with `c2899e2c55b3`.

Runs execute one at a time, each in a new process, on the same arm64 Mac with macOS 15.8.1, using one numerical and one PyTorch CPU thread. Each method keeps its original software environment. Model initialization and fitting are timed together, and full test scoring is timed separately. Imports, process startup, shared input preparation, model saving, and metric evaluation are outside these timers. Peak memory (resident set size) includes runtime libraries, inputs, and stored training state through scoring, and is recorded before the model is saved and reloaded. It therefore describes the memory requirements of these implementations.

Saved-model size uses Python pickle (protocol 4) and includes stored diagnostics and checkpoints. Training-only data loaders and optimizers of CATCH are removed before saving; installed packages and source code are excluded for all methods. Reloaded models reproduce the test scores within the recorded numerical tolerance. CATCH requires replaying its random state for this check because its evaluation masks are stochastic.

Every planned run is reported. There are 129 completed runs and six runs that exceed the time limit: all three HAI seeds of CATCH and of MTAD-GAT. The 3,600-second limit applies to the entire run, including imports and the reload check. A run that exceeds it has unavailable costs and quality; its limit is not a lower bound on fitting time. Partial AUROC at 5% FPR, AP, and AUROC are recomputed from each completed run's stored scores, excluding the first and last 100 test observations. No GPU scores are substituted. Tables S7–S11 report every method and series. The comparison in the main paper is restricted to the four series completed by all methods, and no nine-method five-dataset mean is computed.

E1 has lower fitting and scoring time, peak memory, and saved-model size than both deep methods on each of the four series completed by all methods. Its quality varies by dataset: MTAD-GAT exceeds E1 on the selected SMAP series, and CATCH exceeds E1 on the selected SMD series. On HAI, simpler statistical methods finish faster, and PCA and KMeans exceed E1's partial AUROC. These results show a CPU cost advantage over the two deep implementations for these series. They also document the lower costs of the statistical baselines and the larger HAI resource requirements of DynSphere.

3.8 CPU measurements for four additional methods

A separate study (run record `7326c9f1224e`) adds DAGMM (TAB), OmniAnomaly (TSB-AD), Anomaly Transformer, and TranAD on the same five series and seeds 37–39. The 135 results of the first study are unchanged. Full input arrays and original training budgets are kept, with one numerical and framework CPU thread, runs executed one at a time in new processes, and the same 3,600-second limit per run. Model initialization and fitting are timed together, and full test scoring is timed separately. The transformer implementations construct test windows before scoring, so this preparation is included in their fitting times. Imports, shared input preparation, model saving, and metric evaluation remain outside the timers. Peak memory is measured through the first scoring, before the reload check, and includes runtime libraries and input arrays.

DAGMM is saved as a TensorFlow checkpoint that includes graph and optimizer variables, because its fitted session cannot be pickled. The other added methods save their fitted inference state with Python pickle (protocol 4). Installed packages and source code are excluded from all sizes. Reloaded models reproduce the original scores within the recorded tolerance, with random-state replay where required. Saved-model sizes therefore depend on each implementation's format and do not measure equivalent sets of model parameters.

The study has 45 completed runs and 15 runs that exceed the time limit. DAGMM and TranAD complete all 15 runs. OmniAnomaly completes 12 and exceeds the limit on all three HAI runs. Anomaly Transformer completes the three MSL runs and exceeds the limit on its other 12 runs. Together with the first study, 174 of the 195 planned runs complete and 21 exceed the limit. Runs that exceed the limit have unavailable quality, fitting and scoring time, memory, and model size, and the limit is not a lower bound on fitting time. All 174 completed partial AUROC, AP, and AUROC values are recomputed from their own stored CPU scores, excluding the first and last 100 test observations. We verify the code versions, inputs, configurations, and score checksums of every run. GPU accuracy is never substituted.

Tables S7–S11 report the first nine methods. Tables S12–S16 report all four additional methods, including every run that exceeds the limit. All 13 methods complete every run only on MSL. The main table compares the other 12 methods on the BATADAL, MSL, SMAP, and SMD series they all complete. No 13-method five-dataset mean is computed. On MSL, Anomaly Transformer obtains partial AUROC 0.4977, compared with 0.5106 for E1; its other datasets remain unavailable.

On these four series, E1 has higher mean partial AUROC and lower total fitting and scoring time, peak memory, and saved-model size than CATCH, MTAD-GAT, OmniAnomaly, and TranAD. These aggregate findings allow reversals on individual datasets. DAGMM scores faster than E1, and the statistical baselines have lower costs. Nine methods complete all five series: the four DynSphere configurations, PCA, KMeans, ridge, DAGMM, and TranAD. On these series, E1/DAGMM/TranAD obtain partial AUROC 0.6423/0.5138/0.6156, geometric-mean fitting times 0.5232/1.1112/13.5123 seconds, and scoring times 0.0372/0.0046/2.3126 seconds. DAGMM also fits faster than E1 on HAI. S1 has the highest partial AUROC in this small CPU study. This study provides descriptive quality and cost evidence for the selected series; the primary 59-series accuracy ranking is a separate comparison.

4 Held-out diagnostics for DynSphere and the statistical baselines

This study uses the same data split as the primary comparison. Each MSL, SMAP, and SMD series begins with a training segment free of labeled anomalies, followed by a test segment. We use the first 80% of the training segment for fitting and the remaining 20% for normal calibration, and evaluate only the test segment. BATADAL and HAI provide separate test files. Preprocessing statistics are fitted only on the fitting observations. Every method excludes the first 100 and last 10 test observations per entity, and calibration excludes 10 observations at each boundary. The primary comparison in the main paper instead excludes 100 observations at both test boundaries.

Table S18 reports this diagnostic. It includes the four DynSphere configurations, PCA reconstruction, nearest-centroid distance, and ridge AR(1) prediction. There are 1,281 valid runs over 61 series and three seeds, including the separately reported PowerSystem series. Table S18 displays the remaining 60 series from the five telemetry datasets. One additional SMD series has three labeled anomalies in its training segment, and its 21 runs are excluded. The PowerSystem diagnostic is reported separately because its temporal ordering is unverified. These values and their intervals are kept separate from the primary comparison in the main paper.

Table S7: CPU measurements for BATADAL. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs.

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DynSphere-E1	3/3	0.8070	0.6758	0.9262	0.831	0.0129	347.9	118.6
DynSphere-E4	3/3	0.8202	0.6817	0.9480	0.755	0.0153	330.7	210.5
DynSphere-S1	3/3	0.7726	0.6413	0.9374	0.762	0.0113	322.3	118.6
DynSphere-S4	3/3	0.7856	0.6165	0.9301	0.759	0.0160	326.0	210.5
PCA reconstruction	3/3	0.6988	0.4442	0.8449	0.008	0.0010	307.2	20.7
KMeans distance	3/3	0.6981	0.4432	0.8016	0.099	0.0011	299.8	29.4
Ridge AR(1)	3/3	0.7298	0.4880	0.7946	0.004	0.0011	292.8	15.3
MTAD-GAT (ML4ITS)	3/3	0.6845	0.4449	0.8937	741.583	17.3597	4572.5	1460.3
CATCH	3/3	0.6420	0.3222	0.8173	2464.407	0.5614	3493.9	379537.1

Table S8: CPU measurements for HAI-20.07. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs.

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DynSphere-E1	3/3	0.6563	0.3486	0.6610	208.539	1.0102	3482.1	2965.9
DynSphere-E4	3/3	0.6511	0.3363	0.6593	154.638	1.3511	3487.4	3134.8
DynSphere-S1	3/3	0.6510	0.3485	0.6205	197.968	0.9915	3511.5	2965.8
DynSphere-S4	3/3	0.6562	0.3520	0.6013	157.496	1.4065	3546.2	3134.7
PCA reconstruction	3/3	0.6707	0.3760	0.6092	0.501	0.1055	1325.2	38.6
KMeans distance	3/3	0.6613	0.3539	0.6203	5.126	0.0517	960.2	970.1
Ridge AR(1)	3/3	0.5917	0.2317	0.5667	0.168	0.1242	1450.7	28.2
MTAD-GAT (ML4ITS)	0/3	T	T	T	T	T	T	T
CATCH	0/3	T	T	T	T	T	T	T

Table S9: CPU measurements for MSL. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs.

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DynSphere-E1	3/3	0.5106	0.2484	0.5216	0.020	0.0062	311.5	61.3
DynSphere-E4	3/3	0.5551	0.4058	0.6027	0.030	0.0075	303.7	208.7
DynSphere-S1	3/3	0.4872	0.1775	0.4511	0.019	0.0056	312.5	61.2
DynSphere-S4	3/3	0.4872	0.2011	0.5211	0.031	0.0067	292.5	208.6
PCA reconstruction	3/3	0.4926	0.2166	0.4207	0.001	0.0007	286.7	30.1
KMeans distance	3/3	0.4978	0.1900	0.4898	0.011	0.0008	294.2	4.2
Ridge AR(1)	3/3	0.5005	0.1880	0.4670	0.002	0.0008	284.1	24.6
MTAD-GAT (ML4ITS)	3/3	0.4967	0.1610	0.4032	53.329	10.4947	7183.4	1593.1
CATCH	3/3	0.5010	0.1669	0.3908	172.396	0.3655	3267.3	379544.6

E4 has the highest AUPRC among these methods on BATADAL and SMD, and S4 leads on SMAP. PCA leads on MSL and HAI. E1's AUPRC advantage over KMeans on SMAP is 0.1108, with descriptive paired bootstrap interval $[0.0258, 0.2009]$. Its SMD advantage over ridge prediction is 0.0820, with interval $[0.0306, 0.1286]$. The SMD difference from KMeans has interval $[-0.0092, 0.1006]$. The intervals use 10,000 replicates, average seeds before resampling, and are unadjusted for multiple comparisons.

The PowerSystem experiment fits normal observations from files 1–9, calibrates on normal observations from files 10–12, and evaluates files 13–15. We checked for identical raw feature vectors across partitions and found none in our copy. File boundaries and gaps introduced by selecting normal rows define separate entities. E4 obtains AUPRC 0.8159 and partial AUROC 0.5584, compared with 0.7412/0.5164 for PCA and 0.7317/0.5052 for KMeans. Temporal continuity and scenario independence remain unverified, and the release describes random subsampling [5]. These results therefore show that DynSphere improves on PCA and KMeans when training, calibration, and test data come from disjoint source files, but they do not test temporal motion evidence. Relative to the TSB-AD-M study, this diagnostic changes both the fitting and the evaluation data, so it cannot isolate the effect of excluding the training segment from scoring.

4.1 State and motion evidence

We examine the additive score decomposition using the same fitted E1 model. The state/locality group includes position and concept

Table S10: CPU measurements for SMAP. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs.

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DynSphere-E1	3/3	0.5349	0.0928	0.4433	0.045	0.0152	333.5	35.3
DynSphere-E4	3/3	0.5328	0.0954	0.4426	0.083	0.0190	303.3	69.1
DynSphere-S1	3/3	0.5434	0.1497	0.4605	0.044	0.0149	314.1	35.3
DynSphere-S4	3/3	0.5111	0.0635	0.4314	0.045	0.0185	287.7	69.1
PCA reconstruction	3/3	0.5072	0.0605	0.5003	0.001	0.0010	292.3	8.5
KMeans distance	3/3	0.5029	0.0648	0.4886	0.015	0.0009	286.2	8.0
Ridge AR(1)	3/3	0.5037	0.0668	0.5053	0.002	0.0011	294.2	5.6
MTAD-GAT (ML4ITS)	3/3	0.6082	0.1737	0.5256	84.816	14.4275	2971.3	1286.4
CATCH	3/3	0.5393	0.0872	0.3385	413.652	0.5827	2781.1	379535.0

Table S11: CPU measurements for SMD. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs.

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DynSphere-E1	3/3	0.7025	0.3205	0.8275	0.250	0.0573	363.1	97.0
DynSphere-E4	3/3	0.7025	0.3205	0.8273	0.947	0.0818	395.0	169.7
DynSphere-S1	3/3	0.8697	0.6849	0.9188	0.249	0.0592	363.3	97.0
DynSphere-S4	3/3	0.8365	0.6107	0.9139	0.337	0.0811	359.3	169.7
PCA reconstruction	3/3	0.6660	0.2985	0.7868	0.005	0.0047	292.5	16.6
KMeans distance	3/3	0.6659	0.2986	0.7788	0.070	0.0035	291.5	24.3
Ridge AR(1)	3/3	0.6859	0.3043	0.8280	0.003	0.0052	308.3	12.1
MTAD-GAT (ML4ITS)	3/3	0.5453	0.0899	0.6995	472.455	77.0893	4235.2	1408.9
CATCH	3/3	0.8129	0.4478	0.8990	2237.493	3.1788	3122.3	379539.5

Table S12: Additional CPU measurements for BATADAL. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs. DAGMM is saved as a TensorFlow checkpoint that includes graph and optimizer variables; the other models are saved with Python pickle (protocol 4).

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DAGMM (TAB)	3/3	0.5339	0.1050	0.5949	0.768	0.0014	465.7	309.1
OmniAnomaly (TSB-AD)	3/3	0.6976	0.4289	0.7492	139.723	1.2629	1233.7	163.3
Anomaly Transformer	0/3	T	T	T	T	T	T	T
TranAD	3/3	0.7061	0.4559	0.8389	16.008	0.8266	647.5	1288.8

Table S13: Additional CPU measurements for HAI-20.07. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs. DAGMM is saved as a TensorFlow checkpoint that includes graph and optimizer variables; the other models are saved with Python pickle (protocol 4).

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DAGMM (TAB)	3/3	0.5311	0.0900	0.5735	9.738	0.1333	1355.4	315.3
OmniAnomaly (TSB-AD)	0/3	T	T	T	T	T	T	T
Anomaly Transformer	0/3	T	T	T	T	T	T	T
TranAD	3/3	0.6393	0.3238	0.6269	836.983	79.3732	4108.7	2367.0

distance. The motion/state-speed group includes latent and raw acceleration residuals, speed, and joint state-speed deviation. These groups contain overlapping information because concept distance uses state and velocity and the joint statistic includes both. They provide a diagnostic grouping of score terms.

Each group and their sum use identical propagation radii, $R \in \{0, 8\}$, fitting and calibration observations, and evaluation rows. The fitted model can be reused only with one concept, because removing terms can change the minimum-score concept when there are several concepts. The reconstructed combined score agrees with the reference E1 score within absolute and relative tolerance 10^{-10} for every series and seed. The study contains 183 reference fits and 1,098 score configurations.

Table S14: Additional CPU measurements for MSL. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs. DAGMM is saved as a TensorFlow checkpoint that includes graph and optimizer variables; the other models are saved with Python pickle (protocol 4).

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DAGMM (TAB)	3/3	0.5033	0.2294	0.5434	0.565	0.0010	463.5	313.8
OmniAnomaly (TSB-AD)	3/3	0.5055	0.1850	0.4285	3.138	0.4947	843.5	159.7
Anomaly Transformer	3/3	0.4977	0.1969	0.4976	537.836	215.6306	3676.7	29143.1
TranAD	3/3	0.5336	0.2407	0.4580	1.370	0.4259	568.1	2066.7

Table S15: Additional CPU measurements for SMAP. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs. DAGMM is saved as a TensorFlow checkpoint that includes graph and optimizer variables; the other models are saved with Python pickle (protocol 4).

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DAGMM (TAB)	3/3	0.4969	0.0557	0.4930	0.569	0.0017	452.9	302.2
OmniAnomaly (TSB-AD)	3/3	0.5002	0.0659	0.4965	6.838	2.0650	918.9	162.2
Anomaly Transformer	0/3	T	T	T	T	T	T	T
TranAD	3/3	0.5240	0.1107	0.5014	2.157	0.6578	451.9	467.9

Table S16: Additional CPU measurements for SMD. Quality is the mean over seeds 37–39; costs are medians. T denotes three runs that exceeded the 3,600-second limit, with unavailable quality and costs. DAGMM is saved as a TensorFlow checkpoint that includes graph and optimizer variables; the other models are saved with Python pickle (protocol 4).

Method	Complete	pAUC	AP	AUROC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DAGMM (TAB)	3/3	0.5039	0.0168	0.6468	0.705	0.0066	473.3	307.3
OmniAnomaly (TSB-AD)	3/3	0.6799	0.2571	0.6597	120.629	7.7796	1424.7	228.5
Anomaly Transformer	0/3	T	T	T	T	T	T	T
TranAD	3/3	0.6750	0.2776	0.7899	11.377	3.5993	691.2	1019.1

Table S17: Expanded CPU quality and cost on the four selected BATADAL, MSL, SMAP, and SMD series completed by all 12 listed methods, measured on the same Mac. Quality averages seeds within each series and series equally. Costs are geometric means of per-series medians over three seeds. Peak memory includes the process runtime and inputs through scoring. Size is that of the saved fitted model; DAGMM includes TensorFlow graph and optimizer variables, and the other methods use Python pickle. The preceding tables report HAI, the restricted coverage of Anomaly Transformer, and all 21 runs that exceed the time limit.

Method	pAUC	Fit (s)	Score (s)	Memory (MiB)	Size (KiB)
DynSphere-E1	0.6388	0.117	0.0163	338.5	70.6
DynSphere-E4	0.6526	0.205	0.0205	331.2	150.7
DynSphere-S1	0.6682	0.112	0.0154	327.4	70.6
DynSphere-S4	0.6551	0.137	0.0201	315.1	150.6
PCA reconstruction	0.5911	0.003	0.0013	294.6	17.2
KMeans distance	0.5912	0.032	0.0013	292.9	12.4
Ridge AR(1)	0.6050	0.002	0.0015	294.7	12.7
DAGMM (TAB)	0.5095	0.646	0.0020	463.8	308.1
TranAD	0.6097	4.817	0.9555	582.2	1061.6
OmniAnomaly (TSB-AD)	0.5958	24.523	1.7799	1080.3	176.3
MTAD-GAT (ML4ITS)	0.5837	199.522	21.2165	4509.0	1433.0
CATCH	0.6238	791.879	0.7852	3155.4	379539.0

Table S19 shows that motion/state-speed evidence accounts for most discrimination. With $R = 8$, adding state/locality changes BATADAL AUPRC from 0.6751 to 0.6756 and SMD from 0.4718 to 0.4719. On MSL, the motion group obtains 0.2587, exceeding the combined score's 0.2034. The combined score also falls slightly below the motion group on SMAP. The study provides limited

Table S18: Held-out diagnostic for DynSphere and the statistical baselines, excluding the first 100 and last 10 test observations, reported as AUPRC/standardized partial AUROC for FPR up to 5%. MSL, SMAP, and SMD contain 14, 25, and 19 eligible series. BATADAL and HAI each contain one series. Seeds are averaged within each series before averaging over series.

Method	BATADAL	HAI-20.07	MSL	SMAP	SMD
DynSphere-S4	0.6162/0.7878	0.3520/0.6562	0.1832/0.5544	0.2860/0.6263	0.4600/0.7388
DynSphere-E4	0.6814/0.8217	0.3363/0.6511	0.1706/0.5265	0.2792/0.6241	0.4728/0.7478
DynSphere-S1	0.6407/0.7752	0.3485/0.6510	0.1658/0.5356	0.2448/0.6027	0.4586/0.7409
DynSphere-E1	0.6756/0.8089	0.3486/0.6563	0.2034/0.5426	0.2837/0.6309	0.4719/0.7492
PCA	0.4422/0.6992	0.3760/0.6707	0.3101/0.6151	0.1146/0.5337	0.3544/0.6756
KMeans	0.4414/0.6983	0.3539/0.6613	0.2515/0.5742	0.1728/0.5572	0.4262/0.7147
Ridge AR(1)	0.4871/0.7308	0.2317/0.5917	0.2160/0.5542	0.1750/0.5605	0.3899/0.7051

Table S19: E1 grouped-evidence study, reported as AUPRC. Every group uses the indicated propagation radius and the same held-out series.

Evidence	BATADAL	HAI-20.07	MSL	SMAP	SMD
State/locality ($R = 0$)	0.2306	0.3009	0.1651	0.1468	0.4369
Motion/state-speed ($R = 0$)	0.5144	0.2988	0.1754	0.1674	0.4644
Combined ($R = 0$)	0.5172	0.3023	0.1714	0.1693	0.4654
State/locality ($R = 8$)	0.5866	0.3387	0.1957	0.2146	0.4702
Motion/state-speed ($R = 8$)	0.6751	0.3485	0.2587	0.2857	0.4718
Combined ($R = 8$)	0.6756	0.3486	0.2034	0.2837	0.4719

evidence of a general benefit from combining all terms, so individual terms require justification for each dataset.

Propagation has a substantial effect on BATADAL and SMAP: combined AUPRC increases from 0.5172 to 0.6756 and from 0.1693 to 0.2837, respectively. The SMD change is smaller (0.4654 to 0.4719). Radius zero still uses centered numerical derivatives, so neither setting measures strictly causal detection performance.

5 Additional source-native Exathlon metrics

The source-native Exathlon study is separate from the primary five-dataset comparison and from the TSB-AD Exathlon preparation used in the extension. Table S20 reports all five metrics for the ten configurations and three seeds. The main paper presents AUROC, VUS-PR, and VUS-ROC from the same results.

Table S20: Complete source-native Exathlon comparison. pAUC is standardized partial AUROC for FPR up to 5%. pAUC, AP, and AUROC cover applications 1, 2, 4, and 5 and five test executions. Official VUS 0.0.6 uses maximum buffer 100 on retained contiguous segments that contain both normal and anomalous observations, weighted by retained segment length within each execution. VUS covers applications 2, 4, and 5 and four executions. Seeds are averaged within execution, executions within application, and applications equally. Bold identifies each highest mean.

Method	pAUC	AP	AUROC	VUS-PR	VUS-ROC
KMeans	**0.9301**	**0.8661**	0.9785	0.8942	0.9212
PCA	0.9256	0.8654	0.9694	0.9157	0.9173
Ridge AR(1)	0.9024	0.8509	0.9758	0.9071	0.9272
DynSphere-S4	0.8921	0.8321	0.9810	0.9654	**0.9799**
DynSphere-S1	0.8856	0.8171	**0.9829**	**0.9692**	0.9673
DynSphere-E4	0.8842	0.8087	0.9799	0.9569	0.9547
E1 reduced ($R = 8$)	0.8818	0.8037	0.9809	0.9691	0.9611
DynSphere-E1	0.8818	0.8035	0.9804	0.9608	0.9541
MTAD-GAT (ML4ITS)	0.8451	0.8084	0.9804	0.9279	0.9271
CATCH	0.7963	0.7700	0.9682	0.8908	0.9238

KMeans leads partial AUROC (0.9301), followed by PCA (0.9256) and ridge AR(1) (0.9024). S4 obtains 0.8921, exceeding MTAD-GAT (0.8451) and CATCH (0.7963). KMeans also has the highest AP (0.8661). On applications 1 and 2, for which pilot results were available before the study, S4 obtains partial AUROC 0.9876 and MTAD-GAT obtains 0.9886. On applications 4 and 5, S4 obtains 0.7966, MTAD-GAT obtains 0.7017, and KMeans obtains 0.8926. The reduced E1 score changes the partial AUROC of the complete configuration by less than 0.0001.

Application 1 has no retained contiguous segment that contains both classes. Its complete retained execution contains both classes and is included in the pointwise metrics. This restriction of VUS applies equally to every method. The main paper describes the separate fitting and calibration executions, segment boundaries, preparation choices, and limitations of the application selection.

6 Supporting primary accuracy and PCA dimension sensitivity

6.1 Primary accuracy and temporal metrics

Table S21 repeats the primary 59-series comparison of the main paper and its official VUS summaries. BATADAL, HAI-20.07, MSL, SMAP, and SMD contribute 1, 1, 14, 24, and 19 series, respectively. Each method uses seeds 37–39. Seeds are averaged within each series, series within each dataset, and the five datasets equally. The comparison of all 13 methods contains 2,301 runs. The remaining eligible SMAP series is excluded because DAGMM fails on it at all three seeds; the other methods' results on it appear in the 60-series sensitivity analysis. The SMD series whose training segment contains labeled anomalies remains excluded.

Standardized partial AUROC uses maximum FPR 0.05. Official VUS version 0.0.6 uses a maximum buffer equal to the rounded median length of labeled anomalous events in the series common to all methods. This evaluator setting is shared by all methods, and evaluation labels are never used to fit any method. The primary comparison excludes the first and last 100 test observations. These definitions differ from the held-out diagnostic above, which excludes the first 100 and last 10 test observations.

6.2 PCA95 sensitivity study

The primary configuration remains E1, with Euclidean state, one concept, $q = \min(5, d)$, unchanged weights, and propagation radius 8. We separately evaluate the four DynSphere configurations and eight competitors with a dimension rule fitted only on the fitting data, $q = \min(d, \max(5, q_{95}))$. Here, q_{95} is the smallest dimension that reaches 95% cumulative variance after standardization fitted only on normal fitting observations. Calibration and test observations use the same fitted projection, without whitening. For DynSphere, only the internal PCA dimension changes, and the raw-feature acceleration model is unchanged. For competitors, PCA is an additional input transformation, and architecture and epoch budgets remain fixed. The PCA reconstruction baseline already keeps 95% of the variance and is not tested again. Its original rule has no five-component minimum.

The study records all 2,160 additional runs on 60 eligible series: 2,157 complete and three DAGMM failures on the additional SMAP series. Each of the 12 paired comparisons has the same 59 series and three seeds, giving 177 original/variant pairs. All five metrics are complete on these series, including official VUS-PR and VUS-ROC. Table S22 reports every configuration and both sides of each comparison. The sensitivity study is exploratory: we do not select configurations per dataset or claim an independent model selection.

PCA95 reduces all five metrics for every DynSphere configuration. E1's partial AUROC decreases from 0.6792 to 0.6524, and VUS-PR decreases from 0.4232 to 0.3821. CATCH is the only method whose partial AUROC, AP, and VUS-PR improve: its PCA95 partial AUROC is 0.6838 and VUS-PR is 0.4469. These exceed the corresponding values of the original E1 in this exploratory comparison; the original E1 keeps a higher AP (0.3956 versus 0.3778). CATCH's AUROC and VUS-ROC decrease slightly. Anomaly Transformer's AUROC and VUS-ROC increase slightly from near-chance values. The other competitors lose partial AUROC, AP, and VUS-PR. The variance-retention criterion therefore provides no general ranking benefit, and we keep the original E1 in the primary comparison.

The PCA95 study changes the method or its input transformation. The representation diagnostics below instead apply PCA after fitting, to describe the state produced by the unchanged method. The two experiments answer different questions and are reported separately.

Table S21: Copy of the primary 13-method held-out comparison. Dataset columns and Mean report standardized partial AUROC for FPR up to 5%. Mean, VUS-PR, and VUS-ROC give each dataset equal weight. The table uses the same source as the main paper.

Method	BATADAL	HAI	MSL	SMAP	SMD	*Mean*	*VUS-PR*	*VUS-ROC*
DynSphere-E1	0.8070	0.6563	0.5617	0.6207	**0.7503**	**0.6792**	**0.4232**	0.7779
DynSphere-E4	**0.8202**	0.6511	0.5438	0.6136	0.7490	0.6755	0.4116	0.7736
CATCH	0.6400	0.7069	0.5978	**0.6994**	0.7171	0.6723	0.4216	**0.8460**
DynSphere-S4	0.7856	0.6562	0.5617	0.6161	0.7392	0.6718	0.4047	0.7715
DynSphere-S1	0.7726	0.6510	0.5442	0.5912	0.7417	0.6601	0.3870	0.7712
TranAD	0.7058	0.6407	**0.6373**	0.5867	0.7033	0.6548	0.3559	0.7733
MTAD-GAT (ML4ITS)	0.6843	0.6629	0.5639	0.6295	0.6908	0.6463	0.3474	0.7884
KMeans	0.6981	0.6613	0.5812	0.5583	0.7156	0.6429	0.3540	0.7673
OmniAnomaly (TSB-AD)	0.6976	**0.7478**	0.5233	0.5409	0.6899	0.6399	0.3611	0.7900
PCA	0.6988	0.6707	0.6159	0.5351	0.6766	0.6394	0.3488	0.7563
Ridge AR(1)	0.7298	0.5917	0.5614	0.5556	0.7054	0.6288	0.3300	0.7561
DAGMM (TAB)	0.5339	0.5313	0.5294	0.5469	0.5360	0.5355	0.1383	0.6620
Anomaly Transformer	0.5006	0.5132	0.5051	0.5056	0.5080	0.5065	0.0787	0.5137

Table S22: PCA95 sensitivity on the same 59 primary series and three seeds. Every entry is an equal-dataset mean. Original configurations are unchanged; PCA95 denotes the separately fitted variant with the five-component minimum. pAUC uses FPR up to 5%. AP is average precision.

Method	Configuration	pAUC	AP	AUROC	VUS-PR	VUS-ROC
DynSphere-E1	Original	0.6792	0.3956	0.7322	0.4232	0.7779
DynSphere-E1	PCA95	0.6524	0.3645	0.7233	0.3821	0.7683
KMeans distance	Original	0.6429	0.3345	0.7106	0.3540	0.7673
KMeans distance	PCA95	0.6312	0.3039	0.6807	0.3217	0.7424
Ridge AR(1)	Original	0.6288	0.3029	0.6895	0.3300	0.7561
Ridge AR(1)	PCA95	0.6118	0.2609	0.6612	0.2872	0.7288
DynSphere-E4	Original	0.6755	0.3870	0.7307	0.4116	0.7736
DynSphere-E4	PCA95	0.6528	0.3665	0.7227	0.3831	0.7673
DynSphere-S1	Original	0.6601	0.3687	0.7231	0.3870	0.7712
DynSphere-S1	PCA95	0.6511	0.3564	0.7173	0.3789	0.7697
DynSphere-S4	Original	0.6718	0.3789	0.7224	0.4047	0.7715
DynSphere-S4	PCA95	0.6516	0.3566	0.7190	0.3800	0.7674
DAGMM (TAB)	Original	0.5355	0.1207	0.5921	0.1383	0.6620
DAGMM (TAB)	PCA95	0.5199	0.0999	0.5869	0.1194	0.6533
TranAD	Original	0.6548	0.3514	0.7226	0.3559	0.7733
TranAD	PCA95	0.6293	0.3023	0.6272	0.3157	0.6978
OmniAnomaly (TSB-AD)	Original	0.6399	0.3178	0.7122	0.3611	0.7900
OmniAnomaly (TSB-AD)	PCA95	0.6349	0.3129	0.6927	0.3305	0.7618
MTAD-GAT (ML4ITS)	Original	0.6463	0.3311	0.7427	0.3474	0.7884
MTAD-GAT (ML4ITS)	PCA95	0.6320	0.2980	0.7179	0.3110	0.7657
CATCH	Original	0.6723	0.3647	0.7816	0.4216	0.8460
CATCH	PCA95	0.6838	0.3778	0.7804	0.4469	0.8441
Anomaly Transformer	Original	0.5065	0.0647	0.5035	0.0787	0.5137
Anomaly Transformer	PCA95	0.5043	0.0631	0.5063	0.0779	0.5185

7 Controlled explanations and representations

7.1 Series, procedure, and completion

This study runs on a Linux CPU server and uses the same five series as the CPU cost study, selected from dataset metadata only: BATADAL, HAI-20.07, MSL015, SMAP152, and SMD074. The full series identifiers are

```
015_MSL_id_14_Sensor_tr_575_1st_1250
152_SMAP_id_9_Sensor_tr_2073_1st_5600
074_SMD_id_18_Facility_tr_7174_1st_21230.
```

There are 13 configurations, five series, and seeds 37–39, giving 195 distinct runs. We first ran seed 37 as a pilot, which counts as one of the three seeds. Seeds 38–39 were run only after a technical check of all 65 pilot runs: correct inputs and software versions, finite and repeatable scores, correct reloading of fitted models, valid edits, and complete E1 results on every dataset. Explanation rankings and PCA statistics played no role in this check.

All original normal fitting and calibration data and epoch budgets are unchanged. Each run uses a new process with one CPU and one numerical and framework thread. Fitting, full scoring, and model reloading share a 3,600-second limit, and explanation has a separate 3,600-second limit. An overall limit of 7,230 seconds applies to the explanation process. A subsequent representation step has its own 3,600-second limit and does not affect explanation results or measured explanation time. DAGMM reloads its saved TensorFlow checkpoint instead of refitting; we verify that repeated full scoring reproduces its scores.

The Linux models are fitted separately from the Mac models. Preliminary checks found that four neural implementations produce different scores when refitted on Mac and Linux. The Linux study therefore checks reproducibility only within Linux, by reloading fitted models and repeating scoring with replayed random states. Its results are separate from the Mac replication and from the Mac CPU timing study. No failed, incomplete, or time-limited run is replaced by another run.

The study ends with 163 complete explanation runs and 32 runs that exceed a time limit, with no implementation failure or interrupted run. Table S23 reports all methods and the availability of their representations. OmniAnomaly exceeds the limit on two HAI fits. MTAD-GAT exceeds it on all HAI fits and all MSL explanations. CATCH exceeds it on all BATADAL, HAI, and SMD fits. Anomaly Transformer exceeds it in explanation on BATADAL, MSL, and SMAP, in fitting on HAI, and in fitting or explanation on SMD. A run that exceeds a limit has unavailable measurements under these budgets. Partial results of such runs are stored separately.

7.2 Controlled inputs and common explanations

Before scoring, we choose the chronologically first and last eligible non-overlapping 512-row clean test contexts within one entity. Labels define clean backgrounds and are not available to fitting or attribution. Both selected sensors vary during training and inside rows 240–271, and they are not symmetric in time inside this interval, which the reversal edit requires. Selected starts are 0/2996 for BATADAL, 0/291088 for HAI, 0/1216 for MSL, 14/3015 for SMAP, and 0/21010 for SMD. No model score informs selection.

Each background receives three fixed edits to two sensors over rows 240–271: a mean displacement of two training standard deviations, threefold amplification of residuals, and reversal of the segment. A squared-sine taper makes the displacement smooth at the interval endpoints. The six edited inputs per series are shared by every method and seed. They identify changed inputs on real backgrounds; physical fault causes and the usefulness of explanations to people are outside these annotations.

The explained quantity is the maximum anomaly score of the original method over rows 240–271 of the 512-row context. Sensor groups replace one sensor over rows 128–383; temporal groups replace all sensors in eight consecutive 32-row blocks over the same interval. Replacement values come from the first 256 normal fitting rows; a second reference uses the last 256 rows. Reflection padding lies outside the replaced interval. Every score evaluation replays the fitted random state. The score decrease under one replacement defines its signed attribution ϕ_g.

Table S23: All 195 Linux run outcomes. C, T, U, and N/A denote complete, time limit exceeded, unavailable, and not applicable. Each method has 15 runs. The representation analysis can finish after the explanation step exceeds its limit, because the verified fitted model remains available.

Method	Expl. C	Expl. T	Repr. C	Repr. U	Repr. N/A
DynSphere-E1	15	0	15	0	0
PCA reconstruction	15	0	15	0	0
KMeans distance	15	0	0	0	15
Ridge AR(1)	15	0	0	0	15
DynSphere-E4	15	0	15	0	0
DynSphere-S1	15	0	15	0	0
DynSphere-S4	15	0	15	0	0
DAGMM (TAB)	15	0	0	15	0
TranAD	15	0	15	0	0
OmniAnomaly (TSB-AD)	13	2	13	2	0
MTAD-GAT (ML4ITS)	9	6	12	3	0
CATCH	6	9	6	9	0
Anomaly Transformer	0	15	10	5	0

Sensor and temporal infidelity use 32 independently sampled held-out coalitions containing two to four groups, capped by the number of groups. For measured joint score decreases Δ_j, we compute

$$I = \frac{\frac{1}{32}\sum_{j=1}^{32}(\sum_{g\in C_j}\phi_g - \Delta_j)^2}{\frac{1}{32}\sum_{j=1}^{32}\Delta_j^2}. \quad (1)$$

A zero denominator is reported as unavailable. The same attributions are also evaluated against score decreases under the second reference. This measures how well the attributions approximate the effect of the chosen statistical replacement; whether joint replacements are physically feasible requires separate evidence.

Positive attributions define edited-sensor and edited-time AP and expected precision at two, with ties shared fractionally. Every edit is kept; the subset of edits that increase the score above its clean background is reported separately, with its coverage. Gaussian noise at 0.01 training standard deviations measures the stability of attribution ranks. The score-stable subset satisfies $|\Delta f| \le 0.05|f| + 10^{-8}$. A fixed shuffled attribution serves as a control and receives the same tests. Because the shuffle uses the same rank permutation for every input, it preserves stability correlation and provides no independent stability baseline.

We average the six edited inputs within each series and seed, then seeds within each series, and then datasets equally. A paired comparison on a dataset requires all three seeds and available metrics for both methods. Results on all five datasets are reported separately from comparisons on fewer datasets. A mean over all 13 methods and five datasets is unavailable because some runs exceed the time limit.

On all five datasets, E1 has lower sensor infidelity than PCA and DAGMM. Ridge and TranAD obtain lower values than E1, and E1's sensor and temporal localization AP are below those of several competitors. Every check of E1's component sum and of its radius-8 propagation origins passes (90/90 edited contexts). Its cross-seed sensor rank agreement is 1.00, compared with 0.367 for TranAD; PCA and ridge also obtain 1.00. The other DynSphere configurations improve some sensor diagnostics; all are reported, and E1 was not changed after the evaluation.

Only 48 of the 90 E1 edits increase its score. All 90 scores satisfy the noise-stability condition. Sensor infidelity increases from 0.1780 with the first reference to 0.4589 with the second, which shows sensitivity to the choice of reference. E1's sensor AP exceeds that of its shuffled control (0.2380 versus 0.1223). Explanation takes 3.46 seconds per six inputs on average, compared with 161.60 for TranAD on the same five series. PCA and ridge are faster than E1. These timings are specific to this machine and are kept separate from the Mac quality and cost study.

7.3 Compactness and reproducibility of saved attributions

Cross-seed rank agreement averages all three seed pairs for each edited input. A missing input or metric at any seed makes the series-level value unavailable. We also truncate positive sensor and time attributions to the largest one, two, four, or all groups, using the original held-out joint score decreases. Signed full attributions are a separate diagnostic. Ties follow a fixed group order. This analysis requires no new score evaluations and measures a truncated additive approximation on the saved coalitions.

For E1, sensor infidelity decreases from 0.5357 with one group to 0.4391 with all positive contributions; signed-full infidelity is 0.1780. Temporal attributions yield 0.7013 for every number of positive groups, and signed-full yields 0.7888. The unchanged positive temporal curve therefore says little, and it does not show that any selected temporal group is causally sufficient.

Table S24: Controlled explanations completed on all five datasets and three seeds, for the methods that complete them. I_s, I_t are normalized sensor and time infidelity (lower is better), AP_s, AP_t are AP for localizing the known edits, and $P@2$ is expected precision at two (higher is better). These explanation diagnostics differ from the full-sequence anomaly-ranking AP.

Method	I_s	I_t	AP_s	AP_t	$P@2_s$	$P@2_t$
DynSphere-E1	0.1780	0.7888	0.2380	0.4000	0.1522	0.3786
PCA reconstruction	0.1934	0.5265	0.2912	0.6000	0.2321	0.5500
KMeans distance	0.1259	0.4507	0.2976	0.6000	0.2444	0.5500
Ridge AR(1)	0.0531	0.4688	0.4662	0.4625	0.3861	0.4321
DynSphere-E4	0.2136	0.7803	0.2939	0.3875	0.2259	0.3679
DynSphere-S1	0.1480	0.7498	0.2858	0.4000	0.2102	0.3786
DynSphere-S4	0.1114	0.8474	0.3388	0.3625	0.2691	0.3464
DAGMM (TAB)	0.3294	0.5309	0.3428	0.5042	0.3193	0.4679
TranAD	0.0267	0.4708	0.4969	0.5917	0.3891	0.5548

Table S25: Explanation comparisons on fewer datasets. Each comparator and E1 use the same completed series and all three seeds. F is the number of datasets compared. OmniAnomaly uses BATADAL/MSL/SMAP/SMD, MTAD-GAT uses BATADAL/SMAP/SMD, and CATCH uses MSL/SMAP. Dashes indicate that no completed comparison with Anomaly Transformer is available.

Comparator	F	I_s(E1)	I_s(comp.)	AP_s(E1)	AP_s(comp.)	I_t(E1)	I_t(comp.)
OmniAnomaly (TSB-AD)	4	0.2225	0.2061	0.2621	0.3956	0.7328	3.4224
MTAD-GAT (ML4ITS)	3	0.2828	0.0336	0.2396	0.5560	0.9300	0.5921
CATCH	2	0.2263	0.0505	0.2157	0.4512	0.2532	0.0480
Anomaly Transformer	0	–	–	–	–	–	–

Table S26: Sensor diagnostics on the datasets each method completes with all three seeds. Comparisons between methods require the same listed datasets. I^{rand}, AP^{rand} use the shuffled attribution; I^{donor2} uses the second reference; S_s is rank agreement under noise. F_{seed} is the number of datasets with defined three-seed rank agreement. N_{inc} is the mean number of score-increasing edits per six inputs. Time is the mean explanation time in seconds per six inputs, excluding fitting; it is separate from the Mac timing comparison.

Method	F	I_s^{rand}	AP_s^{rand}	I_s^{donor2}	S_s	F_{seed}	S_{seed}	N_{inc}	Time(s)
DynSphere-E1	5	1.4831	0.1223	0.4589	0.8454	5	1.0000	3.20	3.46
PCA reconstruction	5	1.4972	0.1212	0.6505	0.8437	5	1.0000	3.60	0.33
KMeans distance	5	9072.4678	0.1272	0.5836	0.8250	5	0.9996	3.40	0.67
Ridge AR(1)	5	7.3650	0.1287	0.2036	0.8340	5	1.0000	3.80	0.53
DynSphere-E4	5	1.4616	0.1340	0.7526	0.8262	5	0.8797	3.87	5.14
DynSphere-S1	5	23.3177	0.1270	0.4411	0.8503	5	1.0000	3.40	3.52
DynSphere-S4	5	1.6774	0.1018	0.5264	0.8002	5	0.9786	2.87	5.20
DAGMM (TAB)	5	2.2122	0.1164	67.1479	0.7780	5	0.0491	3.40	0.54
TranAD	5	1.5220	0.1501	0.9142	0.8492	5	0.3667	3.87	161.60
OmniAnomaly (TSB-AD)	4	9961.0756	0.1772	2.6167	0.7849	4	0.9223	2.25	113.61
MTAD-GAT (ML4ITS)	3	1.2045	0.0983	0.3250	0.9649	3	0.7721	4.44	2512.55
CATCH	2	8448.5968	0.0882	1.4182	0.8228	2	0.5104	3.67	222.49
Anomaly Transformer	0	–	–	–	–	0	–	–	–

Table S27: Temporal diagnostics under the second reference, the shuffled control, and noise, with the same definitions as Table S26. Constant rank vectors have no defined cross-seed agreement, so F_{seed} can be smaller than the number of datasets with completed explanations.

Method	F	I_t^{rand}	AP_t^{rand}	I_t^{donor2}	S_t	F_{seed}	S_{seed}
DynSphere-E1	5	1.3126	0.3000	1.3772	1.0000	5	1.0000
PCA reconstruction	5	0.9604	0.3500	52.2679	1.0000	5	1.0000
KMeans distance	5	1.3616	0.4000	0.8659	1.0000	5	1.0000
Ridge AR(1)	5	1.3623	0.3375	0.6386	1.0000	5	1.0000
DynSphere-E4	5	1.3380	0.2875	1.1387	1.0000	5	1.0000
DynSphere-S1	5	1.3953	0.2500	2.0836	0.9034	5	1.0000
DynSphere-S4	5	1.2939	0.2625	4.1351	0.8944	5	0.9982
DAGMM (TAB)	5	1.1635	0.3500	16.6895	1.0000	3	0.2969
TranAD	5	1.0992	0.3778	4.0673	0.9919	5	0.3920
OmniAnomaly (TSB-AD)	4	4.1520	0.4312	4.2323	0.9271	4	0.9950
MTAD-GAT (ML4ITS)	3	1.1055	0.4043	2.0632	0.9879	3	0.8811
CATCH	2	1.2623	0.3553	1.8210	0.9172	2	0.6139
Anomaly Transformer	0	–	–	–	–	0	–

Table S28: Conditional subsets and their coverage. Increase and Stable count the qualifying edited inputs among all completed explanations, including any single completed seed outside three-seed coverage. Conditional AP means require three available seeds per series; F_s, F_t give the number of datasets. Stable-score rank means use the same three-seed requirement. These subsets can differ between methods and are descriptive.

Method	Increase	Stable	F_s	AP_s^{inc}	F_t	AP_t^{inc}	S_s^{stable}	S_t^{stable}
DynSphere-E1	48/90	90/90	4	0.3323	4	0.3438	0.8454	1.0000
PCA reconstruction	54/90	90/90	4	0.3751	4	0.6094	0.8437	1.0000
KMeans distance	51/90	90/90	5	0.3471	5	0.5950	0.8250	1.0000
Ridge AR(1)	57/90	90/90	5	0.5295	5	0.4688	0.8340	1.0000
DynSphere-E4	58/90	90/90	5	0.3427	5	0.3750	0.8262	1.0000
DynSphere-S1	51/90	84/90	4	0.4018	4	0.3438	0.8625	1.0000
DynSphere-S4	43/90	90/90	5	0.4758	5	0.3250	0.8002	0.8944
DAGMM (TAB)	51/90	69/90	5	0.4324	5	0.5192	0.7860	1.0000
TranAD	58/90	90/90	5	0.5245	5	0.5921	0.8492	0.9919
OmniAnomaly (TSB-AD)	30/78	73/78	4	0.5235	4	0.6042	0.7849	0.9271
MTAD-GAT (ML4ITS)	40/54	54/54	3	0.4946	3	0.5312	0.9649	0.9879
CATCH	22/36	31/36	2	0.5080	2	0.7885	0.8436	0.8930
Anomaly Transformer	0/0	0/0	0	–	0	–	–	–

Table S29: Held-out infidelity versus the number of positive attributions kept, for all planned methods, with unavailable results shown. Signed full keeps the original positive and negative contributions. Each row averages the F datasets that the method completes with all three seeds; rows with different datasets require a paired comparison before comparing methods.

Method	Axis	F	1	2	4	All positive	Signed full
DynSphere-E1	sensor	5	0.5357	0.5037	0.4637	0.4391	0.1780
DynSphere-E1	time	5	0.7013	0.7013	0.7013	0.7013	0.7888
PCA reconstruction	sensor	5	0.6387	0.6406	0.6266	0.6273	0.1934
PCA reconstruction	time	5	0.5930	0.5930	0.5930	0.5930	0.5265
KMeans distance	sensor	5	0.6116	0.4797	0.4689	0.4710	0.1259
KMeans distance	time	5	0.5775	0.5775	0.5775	0.5775	0.4507
Ridge AR(1)	sensor	5	0.5644	0.4901	0.4434	0.4451	0.0531
Ridge AR(1)	time	5	0.5389	0.5389	0.5389	0.5389	0.4688
DynSphere-E4	sensor	5	0.6070	0.5817	0.5530	0.5563	0.2136
DynSphere-E4	time	5	0.7146	0.7146	0.7146	0.7146	0.7803
DynSphere-S1	sensor	5	0.7389	0.6933	0.6740	0.6759	0.1480
DynSphere-S1	time	5	0.7604	0.7604	0.7604	0.7604	0.7498
DynSphere-S4	sensor	5	0.5783	0.4840	0.4896	0.5121	0.1114
DynSphere-S4	time	5	0.9346	0.9346	0.9346	0.9346	0.8474
DAGMM (TAB)	sensor	5	0.7364	0.6993	0.6965	0.7541	0.3294
DAGMM (TAB)	time	5	0.7914	0.7914	0.7914	0.7914	0.5309
TranAD	sensor	5	0.3871	0.2906	0.2536	0.2624	0.0267
TranAD	time	5	0.5501	0.5409	0.5409	0.5409	0.4708
OmniAnomaly (TSB-AD)	sensor	4	0.6551	0.4153	0.3243	0.3226	0.2061
OmniAnomaly (TSB-AD)	time	4	0.5088	0.5646	0.6504	0.6686	3.4224
MTAD-GAT (ML4ITS)	sensor	3	0.4939	0.4066	0.3645	0.3672	0.0336
MTAD-GAT (ML4ITS)	time	3	0.6145	0.5647	0.5493	0.5493	0.5921
CATCH	sensor	2	0.5125	0.3589	0.2332	0.2215	0.0505
CATCH	time	2	0.2978	0.2795	0.2793	0.2793	0.0480
Anomaly Transformer	sensor	0	–	–	–	–	–
Anomaly Transformer	time	0	–	–	–	–	–

7.4 Representation PCA and held-out reconstruction

Representation diagnostics reuse the verified fitted models after the explanation step and require no additional fitting. We choose up to 128 evenly spaced eligible 256-row normal fitting contexts and 128 clean held-out contexts, within entity boundaries, with at least 32 in each partition. Contexts may overlap. The same context starts are used for every available method and seed within a series. Each context yields one representation vector, obtained by averaging all activation axes other than the feature axis.

We fixed the extracted layers in advance: the DynSphere PCA state (unchanged for Euclidean geometry and unit-normalized for spherical geometry), the retained coordinates of the PCA reconstruction baseline, MTAD-GAT's final GRU state, OmniAnomaly's posterior mean before the normalizing flow, the Anomaly Transformer encoder, TranAD's final encoder output, and CATCH's channel transformer output. The posterior mean summarizes OmniAnomaly's stochastic inference state. The saved DAGMM model does not expose its compression layer, so DAGMM is unavailable. KMeans and ridge have no comparable learned representation and are not applicable. We did not change the layer choice after observing results. We verify that extracting activations leaves the scores unchanged, that activations are repeatable, and that the random state is restored.

PCA is fitted only to training representations. We report both covariance spectra and correlation spectra after standardizing each coordinate on the training data. EV_k is the cumulative explained variance of the first k principal components, and q_{95} is the smallest count reaching 95%. For normalized eigenvalues p_j, the entropy effective rank is $r_{eff} = \exp(-\sum_j p_j \log p_j)$; its width-normalized version is r_{eff}/D_z. We report the native width D_z and the largest estimable rank $\min(N_{train} - 1, D_z)$ to aid interpretation. The contexts are descriptive samples and support no inference that assumes independent samples.

The training PCA also reconstructs held-out representations at ranks one, two, five, and q_{95}, capped by the native width and the fitted PCA rank. We report one minus the residual squared energy divided by the held-out squared energy around the training center. A zero denominator is reported as unavailable. A separate ridge probe (penalty 1) predicts the standardized raw context means from the standardized representations. Its held-out R^2 measures how well the representation recovers this input summary. Neither test uses held-out observations to fit the transformation or the probe.

E1's first two components explain 83.64% of its representation variance, and its mean covariance q_{95} is 3.00 on all five series. PCA reconstruction and TranAD require 5.40 and 7.53 components on the same series. Paired comparisons on fewer datasets also show a lower absolute dimension for E1 than for MTAD-GAT, CATCH, and Anomaly Transformer. OmniAnomaly has a lower q_{95} on its four series (2.33 versus 2.75). Width-normalized effective rank gives a different comparison, since the wider neural representations use a smaller fraction of their available coordinates. E1's five-component explained variance is 100% by construction. These results describe the covariance of a prescribed low-dimensional state. Anomaly information, semantic explanation, and physical root causes require additional evidence.

The probe on raw context means gives a five-dataset mean R^2 of 0.0462 for E1 and 0.3950 for PCA reconstruction. Several neural probe values are strongly negative, including extreme TranAD values on HAI and MSL. These values do not support a favorable ranking of E1 in information retention. Recomputing the probe in a shared software environment initially gave different values for 13 records. Repeating the computation in each method's original software environment reproduced all 131 complete representation diagnostics within the stated numerical tolerances. No model, stored diagnostic, or tolerance was changed.

Table S30: Context-pooled representation diagnostics on the F datasets each method completes with all three seeds. D_z is the mean native width; q_{95} and q_{95}^{corr} use covariance and correlation PCA, respectively. r_{eff} uses covariance. Methods with unavailable or not-applicable representations are listed in Table S23. Comparisons between methods with different datasets require the paired results in Table S32.

Method	F	D_z	EV_1	EV_2	EV_5	q_{95}	q_{95}^{corr}	r_{eff}	r_{eff}/D_z	Probe R^2
DynSphere-E1	5	5.00	0.6460	0.8364	1.0000	3.00	4.20	2.7082	0.5416	0.0462
PCA reconstruction	5	12.20	0.5736	0.7200	0.9164	5.40	7.60	4.5239	0.3505	0.3950
DynSphere-E4	5	5.00	0.6460	0.8364	1.0000	3.00	4.20	2.7082	0.5416	0.0462
DynSphere-S1	5	5.00	0.7194	0.8633	1.0000	2.60	3.60	2.4239	0.4848	-0.2537
DynSphere-S4	5	5.00	0.7194	0.8633	1.0000	2.60	3.60	2.4239	0.4848	-0.2537
TranAD	5	88.00	0.6387	0.7429	0.8828	7.53	9.87	5.3218	0.0592	-4.93e+18
OmniAnomaly (TSB-AD)	4	8.00	0.7900	0.9384	0.9949	2.33	3.00	1.9972	0.2496	-0.6152
MTAD-GAT (ML4ITS)	4	150.00	0.5569	0.7101	0.8713	9.92	11.17	7.1138	0.0474	-0.5697
CATCH	2	256.00	0.2042	0.3676	0.6993	11.50	12.67	10.8444	0.0424	-0.7889
Anomaly Transformer	3	512.00	0.5420	0.6852	0.8444	12.11	13.89	7.7488	0.0151	-18.3767

Table S31: Context-pooled representation compactness for the methods completed on all five datasets and three seeds. Entries are equal-dataset mean q_{95}. The full representation study has 131 completed diagnostics, 34 unavailable outcomes, and 30 not-applicable cases.

Method	q_{95}
DynSphere-E1	3.00
PCA reconstruction	5.40
TranAD	7.53

Table S32: Paired covariance q_{95} on the same series and all three seeds. Results on all five datasets appear in Table S31; comparisons on fewer datasets are reported here.

Comparator	F	q_{95}(E1)	q_{95}(comp.)
PCA reconstruction	5	3.00	5.40
TranAD	5	3.00	7.53
MTAD-GAT (ML4ITS)	4	2.75	9.92
CATCH	2	2.50	11.50
Anomaly Transformer	3	3.00	12.11
OmniAnomaly (TSB-AD)	4	2.75	2.33

Table S33: Held-out reconstruction of each method's own representation using PCA fitted on training data, at ranks one, two, and five. Cov. and Corr. denote covariance and correlation representations. Ranks are capped at the native width and the available training rank; dashes mark unavailable metrics. Reported results require completion on all three seeds.

Method	F	Cov.1	Cov.2	Cov.5	Corr.1	Corr.2	Corr.5
DynSphere-E1	5	0.2924	0.6699	1.0000	0.4018	0.6492	1.0000
PCA reconstruction	5	0.2051	0.3510	0.7223	0.1190	0.2828	0.6269
DynSphere-E4	5	0.2924	0.6699	1.0000	0.4018	0.6492	1.0000
DynSphere-S1	5	0.4731	0.7484	1.0000	0.3181	0.6422	1.0000
DynSphere-S4	5	0.4731	0.7484	1.0000	0.3181	0.6422	1.0000
TranAD	5	0.1649	0.2094	0.3158	0.1162	0.1603	0.3097
OmniAnomaly (TSB-AD)	4	0.6213	0.8050	0.9542	0.4506	0.7327	0.9501
MTAD-GAT (ML4ITS)	4	0.3855	0.5003	0.6896	0.2808	0.3940	0.6140
CATCH	2	0.1586	0.2866	0.5797	0.1379	0.2520	0.5310
Anomaly Transformer	3	0.2124	0.2990	0.4213	0.1178	0.1911	0.3235

7.5 Complete per-series, per-seed explanation outcomes

The following tables report every series, method, and seed. Entries separated by slashes follow seeds 37, 38, and 39. C denotes complete, T_F a run that exceeded the limit during fitting, full scoring, or reloading, and T_E a run that exceeded the limit during explanation. A dash marks an unavailable value and is never replaced by zero. Partial results from time-limited runs are excluded from complete-series means. All aggregate comparisons in the main paper and this supplement use the stated sets of completed series.

Table S34: BATADAL explanation outcomes and all three seed-specific metrics.

Method	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.1547/0.1547/0.1547	0.3227/0.3227/0.3227	1.4401/1.4401/1.4401	0.2500/0.2500/0.2500
PCA reconstruction	C/C/C	0.0951/0.0951/0.0951	0.2849/0.2849/0.2849	0.3977/0.3977/0.3977	0.5625/0.5625/0.5625
KMeans distance	C/C/C	0.3600/0.3604/0.3579	0.2197/0.2385/0.2339	0.1692/0.1673/0.1696	0.5000/0.5000/0.5000
Ridge AR(1)	C/C/C	0.0829/0.0829/0.0829	0.3396/0.3396/0.3396	0.8075/0.8075/0.8075	0.5000/0.5000/0.5000
DynSphere-E4	C/C/C	0.1799/0.1799/0.1787	0.3170/0.3170/0.3031	1.2144/1.2144/1.2120	0.2500/0.2500/0.2500
DynSphere-S1	C/C/C	0.1506/0.1506/0.1506	0.2849/0.2849/0.2849	1.5716/1.5716/1.5716	0.2500/0.2500/0.2500
DynSphere-S4	C/C/C	0.2093/0.2031/0.2098	0.2162/0.2016/0.2493	1.4161/1.4243/1.4192	0.2500/0.2500/0.2500
DAGMM (TAB)	C/C/C	1.0945/1.1975/0.6062	0.1429/0.1975/0.0586	0.0693/0.1945/0.1142	0.6250/0.4375/0.6250
TranAD	C/C/C	0.0729/0.0819/0.0756	0.4955/0.4955/0.3048	0.0562/0.4228/0.2390	0.5625/0.5625/0.5000
OmniAnomaly (TSB-AD)	C/C/C	0.0712/0.0654/0.0606	0.1755/0.1861/0.1788	0.4056/0.4208/0.4026	0.8750/0.8750/0.8750
MTAD-GAT (ML4ITS)	C/C/C	0.0904/0.0815/0.1121	0.2449/0.2949/0.2602	1.4522/1.4856/1.4332	0.2500/0.2500/0.2500
CATCH	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–
Anomaly Transformer	T_E/T_E/T_E	–/–/–	–/–/–	–/–/–	–/–/–

Table S35: HAI-20.07 explanation outcomes and all three seed-specific metrics.

Method	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.0001/0.0001/0.0001	0.1413/0.1413/0.1413	1.0128/1.0128/1.0128	0.2500/0.2500/0.2500
PCA reconstruction	C/C/C	0.0005/0.0005/0.0005	0.2091/0.2091/0.2091	0.9861/0.9861/0.9861	0.6250/0.6250/0.6250
KMeans distance	C/C/C	0.0000/0.0000/0.0000	0.1418/0.1418/0.1418	0.9830/0.9830/0.9830	0.6250/0.6250/0.6250
Ridge AR(1)	C/C/C	0.0006/0.0006/0.0006	0.0877/0.0877/0.0877	1.0060/1.0060/1.0060	0.3125/0.3125/0.3125
DynSphere-E4	C/C/C	0.0001/0.0001/0.0001	0.1444/0.1444/0.1444	1.2872/1.2871/1.2841	0.2500/0.2500/0.2500
DynSphere-S1	C/C/C	0.0015/0.0015/0.0015	0.1544/0.1544/0.1544	0.9505/0.9505/0.9505	0.2500/0.2500/0.2500
DynSphere-S4	C/C/C	0.0015/0.0015/0.0014	0.1014/0.1001/0.1217	1.2276/1.2433/1.2411	0.2500/0.2500/0.2500
DAGMM (TAB)	C/C/C	0.0713/0.1438/0.0801	0.2734/0.1851/0.0777	0.9799/0.9701/0.9451	0.6250/0.6250/0.6250
TranAD	C/C/C	0.0000/0.0000/0.0000	0.1236/0.1725/0.1398	1.0000/1.0000/1.0000	0.6875/0.5000/0.6875
OmniAnomaly (TSB-AD)	C/T_F/T_F	0.0854/–/–	0.3987/–/–	8.7049/–/–	0.2500/–/–
MTAD-GAT (ML4ITS)	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–
CATCH	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–
Anomaly Transformer	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–

Table S36: MSL015 explanation outcomes and all three seed-specific metrics.

Method	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.0417/0.0417/0.0417	0.3298/0.3298/0.3298	0.1411/0.1411/0.1411	0.6250/0.6250/0.6250
PCA reconstruction	C/C/C	0.3613/0.3613/0.3613	0.5182/0.5182/0.5182	0.2256/0.2256/0.2256	0.5625/0.5625/0.5625
KMeans distance	C/C/C	0.2196/0.2196/0.2196	0.5091/0.5091/0.5091	0.1004/0.1004/0.1004	0.6250/0.6250/0.6250
Ridge AR(1)	C/C/C	0.0006/0.0006/0.0006	0.3684/0.3684/0.3684	0.0932/0.0932/0.0932	0.6250/0.6250/0.6250
DynSphere-E4	C/C/C	0.0643/0.0643/0.0643	0.2495/0.2495/0.2495	0.0591/0.0591/0.0591	0.5625/0.5625/0.5625
DynSphere-S1	C/C/C	0.0000/0.0000/0.0000	0.4210/0.4210/0.4210	0.2360/0.2360/0.2360	0.6250/0.6250/0.6250
DynSphere-S4	C/C/C	0.0179/0.0179/0.0179	0.1692/0.1692/0.1692	0.1382/0.1382/0.1382	0.6250/0.6250/0.6250
DAGMM (TAB)	C/C/C	0.5036/0.0000/0.0743	0.4487/0.6093/0.4765	0.6412/0.8810/0.9385	0.4375/0.5625/0.5000
TranAD	C/C/C	0.0000/0.0001/0.1076	0.6093/0.4735/0.5182	0.1119/0.1423/0.1752	0.8750/0.6875/0.6250
OmniAnomaly (TSB-AD)	C/C/C	0.5487/0.5578/0.5537	0.4535/0.4396/0.4396	0.5628/0.5677/0.5680	0.5681/0.5681/0.5681
MTAD-GAT (ML4ITS)	T_E/T_E/T_E	–/–/–	–/–/–	–/–/–	–/–/–
CATCH	C/C/C	0.0508/0.0252/0.0525	0.4672/0.4952/0.5199	0.0471/0.0760/0.0995	0.5833/0.6667/0.9375
Anomaly Transformer	T_E/T_E/T_E	–/–/–	–/–/–	–/–/–	–/–/–

Table S37: SMAP152 explanation outcomes and all three seed-specific metrics.

Method	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.4110/0.4110/0.4110	0.1017/0.1017/0.1017	0.3653/0.3653/0.3653	0.6250/0.6250/0.6250
PCA reconstruction	C/C/C	0.3944/0.3944/0.3944	0.0800/0.0800/0.0800	0.0985/0.0985/0.0985	0.6250/0.6250/0.6250
KMeans distance	C/C/C	0.0464/0.0464/0.0464	0.0800/0.0800/0.0800	0.1043/0.1043/0.1043	0.6250/0.6250/0.6250
Ridge AR(1)	C/C/C	0.0000/0.0000/0.0000	0.8750/0.8750/0.8750	0.0034/0.0034/0.0034	0.6250/0.6250/0.6250
DynSphere-E4	C/C/C	0.8173/0.0054/0.8173	0.3183/0.7700/0.3183	0.4951/0.0510/0.4951	0.6250/0.6250/0.6250
DynSphere-S1	C/C/C	0.4221/0.4221/0.4221	0.1011/0.1011/0.1011	0.5095/0.5095/0.5095	0.6250/0.6250/0.6250
DynSphere-S4	C/C/C	0.0774/0.0774/0.0774	0.6656/0.6656/0.6656	0.9928/0.9928/0.9928	0.4375/0.4375/0.4375
DAGMM (TAB)	C/C/C	0.0323/0.5360/0.0523	0.8467/0.2333/0.2333	0.7627/0.5578/0.3866	0.3750/0.5000/0.5000
TranAD	C/C/C	0.0155/0.0079/0.0153	0.5833/0.7083/0.9167	0.0010/0.0004/0.0010	0.6250/0.6250/0.6250
OmniAnomaly (TSB-AD)	C/C/C	0.0000/0.0000/0.0000	0.8750/0.8750/0.8750	0.0297/0.0296/0.0296	0.6250/0.6250/0.6250
MTAD-GAT (ML4ITS)	C/C/C	0.0008/0.0004/0.0008	0.8750/0.8750/0.8750	0.0021/0.0012/0.0015	0.8472/0.8819/0.9375
CATCH	C/C/C	0.0316/0.0719/0.0712	0.4718/0.4150/0.3381	0.0200/0.0272/0.0182	0.8750/0.8750/0.7500
Anomaly Transformer	T_E/T_E/T_E	–/–/–	–/–/–	–/–/–	–/–/–

Table S38: SMD074 explanation outcomes and all three seed-specific metrics.

Method	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.2828/0.2828/0.2828	0.2943/0.2943/0.2943	0.9846/0.9846/0.9846	0.2500/0.2500/0.2500
PCA reconstruction	C/C/C	0.1159/0.1159/0.1159	0.3640/0.3640/0.3640	0.9246/0.9246/0.9246	0.6250/0.6250/0.6250
KMeans distance	C/C/C	0.0042/0.0041/0.0041	0.5263/0.5263/0.5263	0.8969/0.8972/0.8972	0.6250/0.6250/0.6250
Ridge AR(1)	C/C/C	0.1816/0.1816/0.1816	0.6605/0.6605/0.6605	0.4342/0.4342/0.4342	0.2500/0.2500/0.2500
DynSphere-E4	C/C/C	0.2774/0.2774/0.2774	0.2943/0.2943/0.2943	0.9956/0.9956/0.9956	0.2500/0.2500/0.2500
DynSphere-S1	C/C/C	0.1659/0.1659/0.1659	0.4675/0.4675/0.4675	0.4813/0.4813/0.4813	0.2500/0.2500/0.2500
DynSphere-S4	C/C/C	0.3164/0.2212/0.2212	0.5327/0.5271/0.5271	0.4506/0.4479/0.4479	0.2500/0.2500/0.2500
DAGMM (TAB)	C/C/C	0.1544/0.0464/0.3487	0.3255/0.5804/0.4536	0.0074/0.4971/0.0188	0.4375/0.4375/0.2500
TranAD	C/C/C	0.0018/0.0173/0.0053	0.5772/0.7556/0.5791	0.9919/0.9891/0.9304	0.5000/0.3125/0.5000
OmniAnomaly (TSB-AD)	C/C/C	0.2184/0.2068/0.1909	0.0873/0.0825/0.0788	11.6330/12.7560/13.6640	0.2500/0.2500/0.2500
MTAD-GAT (ML4ITS)	C/C/C	0.0049/0.0042/0.0069	0.5263/0.5263/0.5263	0.3651/0.2836/0.3047	0.3125/0.5000/0.5000
CATCH	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–
Anomaly Transformer	T_F/T_F/T_E	–/–/–	–/–/–	–/–/–	–/–/–

7.6 Complete representation availability and seed-specific results

The representation tables report all 195 runs, including diagnostics computed after the explanation step exceeded its limit. C denotes complete, U unavailable, and N/A not applicable. Covariance q_{95}, correlation q_{95}^{corr}, EV_2, and the input-mean probe are reported separately. Context starts, fitted models, random states, score checksums, and recomputed metrics were checked against the run record of this study, which has identifier `494c08bcef3a60cc38291444d2b6f95cc550e439d9e8d8132f13b9e3e26dbe07`. The data and the E1 configuration are the same as in the main paper.

Table S39: BATADAL representation outcomes. Slashes follow seeds 37/38/39.

Method	Status	D_z(37/38/39)	q_{95}(37/38/39)	q_{95}^{corr}(37/38/39)	EV_2(37/38/39)	Probe R^2(37/38/39)
DynSphere-E1	C/C/C	5/5/5	4/4/4	5/5/5	0.6727/0.6727/0.6727	0.0526/0.0526/0.0526
PCA reconstruction	C/C/C	12/12/12	8/8/8	9/9/9	0.5114/0.5114/0.5114	0.4377/0.4377/0.4377
KMeans distance	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
Ridge AR(1)	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
DynSphere-E4	C/C/C	5/5/5	4/4/4	5/5/5	0.6727/0.6727/0.6727	0.0526/0.0526/0.0526
DynSphere-S1	C/C/C	5/5/5	5/5/5	5/5/5	0.6848/0.6848/0.6848	-0.1113/-0.1113/-0.1113
DynSphere-S4	C/C/C	5/5/5	5/5/5	5/5/5	0.6848/0.6848/0.6848	-0.1113/-0.1113/-0.1113
DAGMM (TAB)	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
TranAD	C/C/C	86/86/86	18/18/19	22/22/22	0.4059/0.3694/0.4211	-1.5726/-1.3274/-1.3331
OmniAnomaly (TSB-AD)	C/C/C	8/8/8	2/2/2	3/2/2	0.9794/0.9675/0.9838	-0.5213/-0.5319/-0.5055
MTAD-GAT (ML4ITS)	C/C/C	150/150/150	29/29/27	31/31/29	0.2929/0.3389/0.3371	-1.2804/-1.6194/-1.3077
CATCH	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
Anomaly Transformer	C/C/C	512/512/512	30/30/23	32/32/28	0.3332/0.3893/0.5744	-6.0172/-5.2458/-5.3830

Table S40: HAI-20.07 representation outcomes. Slashes follow seeds 37/38/39.

Method	Status	D_z(37/38/39)	q_{95}(37/38/39)	q_{95}^{corr}(37/38/39)	EV_2(37/38/39)	Probe R^2(37/38/39)
DynSphere-E1	C/C/C	5/5/5	4/4/4	5/5/5	0.6721/0.6721/0.6721	-0.9376/-0.9376/-0.9376
PCA reconstruction	C/C/C	19/19/19	9/9/9	12/12/12	0.4721/0.4721/0.4721	-0.8664/-0.8664/-0.8664
KMeans distance	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
Ridge AR(1)	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
DynSphere-E4	C/C/C	5/5/5	4/4/4	5/5/5	0.6721/0.6721/0.6721	-0.9376/-0.9376/-0.9376
DynSphere-S1	C/C/C	5/5/5	4/4/4	5/5/5	0.6619/0.6619/0.6619	-1.0289/-1.0289/-1.0289
DynSphere-S4	C/C/C	5/5/5	4/4/4	5/5/5	0.6619/0.6619/0.6619	-1.0289/-1.0289/-1.0289
DAGMM (TAB)	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
TranAD	C/C/C	118/118/118	10/10/10	11/11/11	0.5373/0.5452/0.5472	-4.46e+19/-1.78e+19/-1.16e+19
OmniAnomaly (TSB-AD)	C/U/U	8/–/–	3/–/–	3/–/–	0.9430/–/–	-1.1458/–/–
MTAD-GAT (ML4ITS)	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
CATCH	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
Anomaly Transformer	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–

Table S41: MSL015 representation outcomes. Slashes follow seeds 37/38/39.

Method	Status	D_z(37/38/39)	q_{95}(37/38/39)	q_{95}^{corr}(37/38/39)	EV_2(37/38/39)	Probe R^2(37/38/39)
DynSphere-E1	C/C/C	5/5/5	2/2/2	3/3/3	0.9656/0.9656/0.9656	0.4513/0.4513/0.4513
PCA reconstruction	C/C/C	9/9/9	3/3/3	4/4/4	0.9198/0.9198/0.9198	0.7348/0.7348/0.7348
KMeans distance	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
Ridge AR(1)	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
DynSphere-E4	C/C/C	5/5/5	2/2/2	3/3/3	0.9656/0.9656/0.9656	0.4513/0.4513/0.4513
DynSphere-S1	C/C/C	5/5/5	1/1/1	3/3/3	0.9910/0.9910/0.9910	0.2532/0.2532/0.2532
DynSphere-S4	C/C/C	5/5/5	1/1/1	3/3/3	0.9910/0.9910/0.9910	0.2532/0.2532/0.2532
DAGMM (TAB)	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
TranAD	C/C/C	110/110/110	3/4/4	6/6/6	0.9176/0.9014/0.9210	-2.50e+14/-7.85e+14/-3.47e+13
OmniAnomaly (TSB-AD)	C/C/C	8/8/8	2/1/1	3/2/2	0.9930/0.9921/0.9973	-0.6578/-0.7001/-0.4675
MTAD-GAT (ML4ITS)	C/C/C	150/150/150	1/1/1	3/2/2	0.9925/0.9952/0.9942	-0.7233/-0.7062/-0.4491
CATCH	C/C/C	256/256/256	10/11/11	12/12/13	0.3861/0.3398/0.3955	-0.6423/-0.6737/-0.5781
Anomaly Transformer	C/C/C	512/512/512	2/2/2	3/3/3	0.9551/0.9562/0.9562	-11.0372/-8.7805/-8.2836

Table S42: SMAP152 representation outcomes. Slashes follow seeds 37/38/39.

Method	Status	D_z(37/38/39)	q_{95}(37/38/39)	q_{95}^{corr}(37/38/39)	EV_2(37/38/39)	Probe R^2(37/38/39)
DynSphere-E1	C/C/C	5/5/5	3/3/3	3/3/3	0.8804/0.8804/0.8804	0.7271/0.7271/0.7271
PCA reconstruction	C/C/C	10/10/10	4/4/4	5/5/5	0.7628/0.7628/0.7628	0.8769/0.8769/0.8769
KMeans distance	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
Ridge AR(1)	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
DynSphere-E4	C/C/C	5/5/5	3/3/3	3/3/3	0.8804/0.8804/0.8804	0.7271/0.7271/0.7271
DynSphere-S1	C/C/C	5/5/5	1/1/1	1/1/1	0.9974/0.9974/0.9974	0.0709/0.0709/0.0709
DynSphere-S4	C/C/C	5/5/5	1/1/1	1/1/1	0.9974/0.9974/0.9974	0.0709/0.0709/0.0709
DAGMM (TAB)	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
TranAD	C/C/C	50/50/50	2/2/2	5/6/5	0.9660/0.9556/0.9565	0.6314/0.7199/0.3599
OmniAnomaly (TSB-AD)	C/C/C	8/8/8	3/3/2	4/4/3	0.8817/0.8299/0.9595	-0.5296/-1.3170/-0.7726
MTAD-GAT (ML4ITS)	C/C/C	150/150/150	4/4/5	5/6/6	0.8876/0.8334/0.7617	0.4034/0.3513/-0.1565
CATCH	C/C/C	256/256/256	12/12/13	13/13/13	0.3526/0.3960/0.3355	-0.7039/-0.8868/-1.2490
Anomaly Transformer	C/C/C	512/512/512	6/7/7	8/8/8	0.6852/0.6611/0.6561	-51.8211/-46.4254/-22.3964

Table S43: SMD074 representation outcomes. Slashes follow seeds 37/38/39.

Method	Status	D_z(37/38/39)	q_{95}(37/38/39)	q_{95}^{corr}(37/38/39)	EV_2(37/38/39)	Probe R^2(37/38/39)
DynSphere-E1	C/C/C	5/5/5	2/2/2	5/5/5	0.9914/0.9914/0.9914	-0.0625/-0.0625/-0.0625
PCA reconstruction	C/C/C	11/11/11	3/3/3	8/8/8	0.9341/0.9341/0.9341	0.7918/0.7918/0.7918
KMeans distance	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
Ridge AR(1)	N/A/N/A/N/A	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
DynSphere-E4	C/C/C	5/5/5	2/2/2	5/5/5	0.9914/0.9914/0.9914	-0.0625/-0.0625/-0.0625
DynSphere-S1	C/C/C	5/5/5	2/2/2	4/4/4	0.9812/0.9812/0.9812	-0.4524/-0.4524/-0.4524
DynSphere-S4	C/C/C	5/5/5	2/2/2	4/4/4	0.9812/0.9812/0.9812	-0.4524/-0.4524/-0.4524
DAGMM (TAB)	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
TranAD	C/C/C	76/76/76	4/4/3	5/5/5	0.8855/0.8883/0.9252	-1.21e+06/-7.39e+05/-6.36e+05
OmniAnomaly (TSB-AD)	C/C/C	8/8/8	5/3/2	6/3/2	0.7696/0.9464/0.9606	-0.4599/-0.4595/-0.4597
MTAD-GAT (ML4ITS)	C/C/C	150/150/150	6/6/6	6/7/6	0.7178/0.6826/0.6868	-0.4485/-0.4478/-0.4522
CATCH	U/U/U	–/–/–	–/–/–	–/–/–	–/–/–	–/–/–
Anomaly Transformer	U/U/C	–/–/512	–/–/8	–/–/10	–/–/0.6822	–/–/-0.4400

8 Mac CPU replication of the explanation study

8.1 Procedure, completion, and validation

We repeat the explanation study of Section 7 on a Mac CPU and report it separately from the Linux study. Both evaluate the same 13 configurations, five series selected from metadata, seeds 37–39, and six controlled edits per series. The fitting data, configurations, replacement references, explained scores, and explanation definitions are unchanged. Each run uses one numerical and one framework thread, with 3,600-second limits for fitting, full scoring, and reloading and for explanation, and an overall limit of 7,230 seconds per run. DAGMM reloads the same saved checkpoints as in the Linux study, so its results are not an independent new fit.

All 195 Mac runs are recorded: 168 complete and 27 that exceed a time limit, with no failed or interrupted runs. As on Linux, seed 37 served as the pilot and counts as one of the three seeds, and seeds 38–39 followed the same technical check. The run scheduler was stopped once and restarted after we confirmed that no run was still active. One completed CATCH run was recovered from its saved record without refitting, and only runs that had not yet started were executed afterwards. The records of the 185 runs completed before the restart are unchanged, and no finished run was repeated or replaced.

The final check verifies the run record, its 5,213 file checksums, every run, the saved inputs, the fitted models, the random states, and the checksums of all full scores and edited inputs. The stored explanation metrics are recomputed for all 1,008 completed edited contexts. Partial results of time-limited runs are stored and checked but excluded from complete-input means. Every check of the component sum and of the radius-8 propagation origins passes for all 90 edited contexts of each DynSphere configuration. The identifier of the Mac run record is `3ea9373454429f834f8604ef997cf193583edc3a37b75483f190b32244fc09e9`.

Table S44: Completion for every configuration. Dataset counts follow BATADAL/HAI/MSL/SMAP/SMD, with three planned seeds per dataset. C and T denote completed runs and runs that exceed a time limit. All 195 runs per machine are reported, and none failed or was interrupted.

Configuration	Mac C by dataset	Linux C by dataset	Mac T	Linux T
DynSphere-E1	3/3/3/3/3	3/3/3/3/3	0	0
PCA reconstruction	3/3/3/3/3	3/3/3/3/3	0	0
KMeans distance	3/3/3/3/3	3/3/3/3/3	0	0
Ridge AR(1)	3/3/3/3/3	3/3/3/3/3	0	0
DynSphere-E4	3/3/3/3/3	3/3/3/3/3	0	0
DynSphere-S1	3/3/3/3/3	3/3/3/3/3	0	0
DynSphere-S4	3/3/3/3/3	3/3/3/3/3	0	0
DAGMM (TAB)	3/3/3/3/3	3/3/3/3/3	0	0
TranAD	3/3/3/3/3	3/3/3/3/3	0	0
OmniAnomaly (TSB-AD)	3/0/3/3/3	3/1/3/3/3	3	2
MTAD-GAT (ML4ITS)	3/0/0/3/3	3/0/0/3/3	6	6
CATCH	3/0/3/3/3	0/0/3/3/0	3	9
Anomaly Transformer	0/0/0/0/0	0/0/0/0/0	15	15

8.2 Comparisons on the same completed series

The two machines use separately fitted models and separate runs. Table S45 compares each configuration only on series with all three seeds and available metrics on both machines. The four DynSphere configurations, PCA, KMeans, ridge, DAGMM, and TranAD complete all five datasets. OmniAnomaly uses BATADAL/MSL/SMAP/SMD, MTAD-GAT uses BATADAL/SMAP/SMD, and CATCH uses MSL/SMAP. CATCH also completes BATADAL and SMD on the Mac; these series appear in its per-series tables but not in the cross-machine comparison. A single completed Linux OmniAnomaly HAI seed does not provide three-seed coverage. Anomaly Transformer completes no explanation run, so a mean over all 13 methods is unavailable. Time limits determine which measurements are available and do not rank explanation quality.

E1's sensor infidelity is 0.1782 on the Mac and 0.1780 on Linux, and its sensor localization AP is 0.2380 on both machines. The ordering of sensor infidelity relative to PCA, KMeans, ridge, DAGMM, and TranAD is unchanged. E1 has lower sensor infidelity than PCA and DAGMM; KMeans, ridge, and TranAD obtain lower values than E1. All five comparators have higher sensor and temporal localization AP on all five datasets. The other DynSphere configurations are reported separately, and E1 was not changed after the evaluation.

8.3 Reproducibility, controls, and explanation cost

Table S46 reports the E1 diagnostics on both machines. Cross-seed sensor attribution-rank agreement is 1.00 for E1, PCA, and ridge; TranAD obtains 0.3515 on the Mac and 0.3667 on Linux. Only 48 of E1's 90 edited inputs increase its score. Conditional localization means have three-seed coverage on BATADAL/HAI/MSL/SMD; SMAP has no qualifying inputs. Conditional subsets differ between methods, so comparing them requires attention to which inputs they include. All 90 E1 inputs satisfy the score-stability condition, and the score-stable subset has the same noise-rank means as the full E1 set. The shuffled control uses the same rank permutation for every input, which preserves correlation and provides no independent stability baseline.

E1's Mac sensor infidelity increases from 0.1782 with the first reference to 0.4582 with the second, close to the Linux values of 0.1780 and 0.4589. Its Mac sensor AP exceeds that of the shuffled control (0.2380 versus 0.1223). Keeping the one, two, four, or all largest positive sensor

Table S45: Mac and Linux explanation means on the same series. Each paired entry is Mac/Linux (M/L), and F is the number of datasets completed with all three seeds on both machines. Datasets receive equal weight after averaging the six inputs per series and seed and then the seeds within each series. I_s, I_t denote sensor and time held-out infidelity (lower is better); AP_s, AP_t denote AP for localizing the controlled edits (higher is better). Results on different datasets require separate interpretation. Dashes mark unavailable results.

Configuration	F	I_s (M/L)	I_t (M/L)	AP_s (M/L)	AP_t (M/L)
DynSphere-E1	5	0.1782/0.1780	0.7895/0.7888	0.2380/0.2380	0.4000/0.4000
PCA reconstruction	5	0.1934/0.1934	0.5265/0.5265	0.2912/0.2912	0.6000/0.6000
KMeans distance	5	0.1259/0.1259	0.4507/0.4507	0.2976/0.2976	0.6000/0.6000
Ridge AR(1)	5	0.0531/0.0531	0.4688/0.4688	0.4662/0.4662	0.4625/0.4625
DynSphere-E4	5	0.2137/0.2136	0.7813/0.7803	0.2937/0.2939	0.3875/0.3875
DynSphere-S1	5	0.1480/0.1480	0.7508/0.7498	0.2858/0.2858	0.4000/0.4000
DynSphere-S4	5	0.1116/0.1114	0.8481/0.8474	0.3416/0.3388	0.3625/0.3625
DAGMM (TAB)	5	0.3294/0.3294	0.5310/0.5309	0.3428/0.3428	0.5042/0.5042
TranAD	5	0.0268/0.0267	0.4705/0.4708	0.4971/0.4969	0.5833/0.5917
OmniAnomaly (TSB-AD)	4	0.2061/0.2061	3.4247/3.4224	0.3952/0.3956	0.5795/0.5795
MTAD-GAT (ML4ITS)	3	0.0452/0.0336	0.5809/0.5921	0.5700/0.5560	0.5185/0.5255
CATCH	2	0.0475/0.0505	0.0462/0.0480	0.4121/0.4512	0.7789/0.7812
Anomaly Transformer	0	–/–	–/–	–/–	–/–

Table S46: E1 replication diagnostics. Means use all five datasets and three seeds unless $F = 4$ is stated. The score-increasing subset uses BATADAL/HAI/MSL/SMD. Counts refer to all 90 completed E1 inputs. Explanation time excludes fitting and is descriptive for each machine.

Diagnostic	Mac	Linux
Sensor precision at two	0.1522	0.1522
Temporal precision at two	0.3786	0.3786
Sensor noise rank agreement	0.8448	0.8454
Temporal noise rank agreement	1.0000	1.0000
Sensor second-donor infidelity	0.4582	0.4589
Temporal second-donor infidelity	1.3784	1.3772
Sensor shuffled-control infidelity	1.4833	1.4831
Temporal shuffled-control infidelity	1.3125	1.3126
Sensor shuffled-control AP	0.1223	0.1223
Temporal shuffled-control AP	0.3000	0.3000
Score-increasing sensor AP ($F = 4$)	0.3323	0.3323
Score-increasing temporal AP ($F = 4$)	0.3438	0.3438
Explanation seconds per six inputs	2.9680	3.4619
Sensor cross-seed rank agreement	1.0000	1.0000
Temporal cross-seed rank agreement	1.0000	1.0000
Score-increasing inputs	48/90	48/90
Noise-score-stable inputs	90/90	90/90
Native accounting and origin checks	90/90	90/90

attributions yields Mac infidelity 0.5358, 0.5032, 0.4638, and 0.4392, respectively; the corresponding Linux values are 0.5357, 0.5037, 0.4637, and 0.4391. The signed-full Mac value is 0.1782. The positive temporal curve remains 0.7013 for every number of groups on both machines, with signed-full values 0.7895 (Mac) and 0.7888 (Linux). This unchanged positive curve says little and does not show that any temporal group is causally sufficient. These diagnostics use the saved attributions and require no new score evaluations or fitting.

On all five datasets, Mac explanation time averages 2.97 seconds for E1, 164.55 for TranAD, 0.37 for PCA, and 0.48 for ridge. Table S47 reports every planned configuration and unavailable comparison. Explanation times are specific to each machine and are descriptive; ratios between machines do not establish a speedup. The CPU fitting and scoring study remains the paper's evidence on run time.

The Mac replication confirms reproducible score accounting and the localization and reference-sensitivity limitations observed on Linux. It has no representation-extraction step, so the Linux representation PCA results in Section 7 remain separate, with their qualifications on native width, pooling, and coverage. Physical fault causes, analyst decision quality, and general comparative interpretability remain unevaluated.

8.4 Complete Mac per-series and per-seed outcomes

The following tables report all 195 Mac runs. Entries separated by slashes follow seeds 37/38/39. C denotes complete, T_F a run that exceeded the limit during fitting, full scoring, or reloading, and T_E a run that exceeded the limit during explanation. A dash marks an unavailable metric. Partial results from time-limited runs are excluded from complete-series means. These tables are reported separately from the Linux per-series results.

Table S47: Mean explanation time in seconds per six inputs, excluding fitting, on the same completed datasets for each Mac/Linux pair. F follows Table S45. These measurements are specific to each machine and separate from the Mac CPU quality and cost study.

Configuration	F	Mac seconds	Linux seconds
DynSphere-E1	5	2.9680	3.4619
PCA reconstruction	5	0.3676	0.3271
KMeans distance	5	0.5574	0.6712
Ridge AR(1)	5	0.4822	0.5255
DynSphere-E4	5	4.0930	5.1397
DynSphere-S1	5	2.9623	3.5223
DynSphere-S4	5	4.1114	5.2021
DAGMM (TAB)	5	0.5558	0.5350
TranAD	5	164.5469	161.6048
OmniAnomaly (TSB-AD)	4	175.4461	113.6071
MTAD-GAT (ML4ITS)	3	2029.0716	2512.5457
CATCH	2	130.4647	222.4921
Anomaly Transformer	0	–	–

Table S48: Mac BATADAL explanation outcomes and seed-specific metrics.

Configuration	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.1546/0.1546/0.1546	0.3227/0.3227/0.3227	1.4396/1.4396/1.4396	0.2500/0.2500/0.2500
PCA reconstruction	C/C/C	0.0951/0.0951/0.0951	0.2849/0.2849/0.2849	0.3977/0.3977/0.3977	0.5625/0.5625/0.5625
KMeans distance	C/C/C	0.3600/0.3604/0.3579	0.2197/0.2385/0.2339	0.1692/0.1673/0.1696	0.5000/0.5000/0.5000
Ridge AR(1)	C/C/C	0.0829/0.0829/0.0829	0.3396/0.3396/0.3396	0.8075/0.8075/0.8075	0.5000/0.5000/0.5000
DynSphere-E4	C/C/C	0.1807/0.1807/0.1785	0.3170/0.3170/0.3031	1.2226/1.2226/1.2093	0.2500/0.2500/0.2500
DynSphere-S1	C/C/C	0.1504/0.1504/0.1504	0.2849/0.2849/0.2849	1.5705/1.5705/1.5705	0.2500/0.2500/0.2500
DynSphere-S4	C/C/C	0.2067/0.2052/0.2109	0.2293/0.2016/0.2671	1.4145/1.4237/1.4199	0.2500/0.2500/0.2500
DAGMM (TAB)	C/C/C	1.0945/1.1975/0.6062	0.1429/0.1975/0.0586	0.0693/0.1945/0.1142	0.6250/0.4375/0.6250
TranAD	C/C/C	0.0799/0.0811/0.0747	0.4955/0.4955/0.2938	0.0732/0.4222/0.2253	0.5625/0.5625/0.5625
OmniAnomaly (TSB-AD)	C/C/C	0.0712/0.0652/0.0606	0.1755/0.1861/0.1755	0.4054/0.4209/0.4023	0.8750/0.8750/0.8750
MTAD-GAT (ML4ITS)	C/C/C	0.1004/0.1065/0.1714	0.3366/0.3366/0.2171	1.2854/1.4401/1.3197	0.2500/0.2500/0.2500
CATCH	C/C/C	0.1840/0.5699/0.2444	0.2193/0.2455/0.3726	1.0026/0.4774/0.8943	0.4375/0.2500/0.3750
Anomaly Transformer	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–

Table S49: Mac HAI-20.07 explanation outcomes and seed-specific metrics.

Configuration	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.0001/0.0001/0.0001	0.1413/0.1413/0.1413	1.0172/1.0172/1.0172	0.2500/0.2500/0.2500
PCA reconstruction	C/C/C	0.0005/0.0005/0.0005	0.2091/0.2091/0.2091	0.9861/0.9861/0.9861	0.6250/0.6250/0.6250
KMeans distance	C/C/C	0.0000/0.0000/0.0000	0.1418/0.1418/0.1418	0.9830/0.9830/0.9830	0.6250/0.6250/0.6250
Ridge AR(1)	C/C/C	0.0006/0.0006/0.0006	0.0877/0.0877/0.0877	1.0060/1.0060/1.0060	0.3125/0.3125/0.3125
DynSphere-E4	C/C/C	0.0001/0.0001/0.0001	0.1444/0.1444/0.1444	1.2847/1.2879/1.2837	0.2500/0.2500/0.2500
DynSphere-S1	C/C/C	0.0015/0.0015/0.0015	0.1544/0.1544/0.1544	0.9568/0.9568/0.9568	0.2500/0.2500/0.2500
DynSphere-S4	C/C/C	0.0012/0.0014/0.0014	0.1126/0.0998/0.1228	1.2377/1.2424/1.2419	0.2500/0.2500/0.2500
DAGMM (TAB)	C/C/C	0.0713/0.1438/0.0801	0.2734/0.1851/0.0777	0.9799/0.9701/0.9451	0.6250/0.6250/0.6250
TranAD	C/C/C	0.0000/0.0000/0.0000	0.1245/0.1754/0.1398	1.0000/1.0000/1.0000	0.5000/0.6875/0.5000
OmniAnomaly (TSB-AD)	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–
MTAD-GAT (ML4ITS)	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–
CATCH	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–
Anomaly Transformer	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–

Table S50: Mac MSL015 explanation outcomes and seed-specific metrics.

Configuration	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.0417/0.0417/0.0417	0.3298/0.3298/0.3298	0.1411/0.1411/0.1411	0.6250/0.6250/0.6250
PCA reconstruction	C/C/C	0.3613/0.3613/0.3613	0.5182/0.5182/0.5182	0.2256/0.2256/0.2256	0.5625/0.5625/0.5625
KMeans distance	C/C/C	0.2196/0.2196/0.2196	0.5091/0.5091/0.5091	0.1004/0.1004/0.1004	0.6250/0.6250/0.6250
Ridge AR(1)	C/C/C	0.0006/0.0006/0.0006	0.3684/0.3684/0.3684	0.0932/0.0932/0.0932	0.6250/0.6250/0.6250
DynSphere-E4	C/C/C	0.0643/0.0643/0.0643	0.2495/0.2495/0.2495	0.0591/0.0591/0.0591	0.5625/0.5625/0.5625
DynSphere-S1	C/C/C	0.0000/0.0000/0.0000	0.4210/0.4210/0.4210	0.2360/0.2360/0.2360	0.6250/0.6250/0.6250
DynSphere-S4	C/C/C	0.0179/0.0179/0.0179	0.1692/0.1692/0.1692	0.1382/0.1382/0.1382	0.6250/0.6250/0.6250
DAGMM (TAB)	C/C/C	0.5036/0.0000/0.0743	0.4487/0.6093/0.4765	0.6412/0.8810/0.9385	0.4375/0.5625/0.5000
TranAD	C/C/C	0.0000/0.0003/0.1046	0.6093/0.4735/0.5182	0.1101/0.1398/0.1763	0.8750/0.6875/0.6250
OmniAnomaly (TSB-AD)	C/C/C	0.5487/0.5578/0.5537	0.4535/0.4396/0.4396	0.5628/0.5677/0.5680	0.5681/0.5681/0.5681
MTAD-GAT (ML4ITS)	T_E/T_E/T_E	–/–/–	–/–/–	–/–/–	–/–/–
CATCH	C/C/C	0.0599/0.0197/0.0558	0.4369/0.4869/0.4922	0.0480/0.0681/0.0912	0.6597/0.6806/0.8264
Anomaly Transformer	T_E/T_E/T_E	–/–/–	–/–/–	–/–/–	–/–/–

Table S51: Mac SMAP152 explanation outcomes and seed-specific metrics.

Configuration	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.4110/0.4110/0.4110	0.1017/0.1017/0.1017	0.3653/0.3653/0.3653	0.6250/0.6250/0.6250
PCA reconstruction	C/C/C	0.3944/0.3944/0.3944	0.0800/0.0800/0.0800	0.0985/0.0985/0.0985	0.6250/0.6250/0.6250
KMeans distance	C/C/C	0.0464/0.0464/0.0464	0.0800/0.0800/0.0800	0.1043/0.1043/0.1043	0.6250/0.6250/0.6250
Ridge AR(1)	C/C/C	0.0000/0.0000/0.0000	0.8750/0.8750/0.8750	0.0034/0.0034/0.0034	0.6250/0.6250/0.6250
DynSphere-E4	C/C/C	0.8173/0.0054/0.8173	0.3183/0.7700/0.3183	0.4951/0.0510/0.4951	0.6250/0.6250/0.6250
DynSphere-S1	C/C/C	0.4219/0.4219/0.4219	0.1011/0.1011/0.1011	0.5094/0.5094/0.5094	0.6250/0.6250/0.6250
DynSphere-S4	C/C/C	0.0774/0.0774/0.0774	0.6656/0.6656/0.6656	0.9928/0.9928/0.9928	0.4375/0.4375/0.4375
DAGMM (TAB)	C/C/C	0.0323/0.5360/0.0523	0.8467/0.2333/0.2333	0.7627/0.5578/0.3866	0.3750/0.5000/0.5000
TranAD	C/C/C	0.0146/0.0083/0.0153	0.5833/0.7083/0.9167	0.0010/0.0004/0.0010	0.6250/0.6250/0.6250
OmniAnomaly (TSB-AD)	C/C/C	0.0000/0.0000/0.0000	0.8750/0.8750/0.8750	0.0297/0.0296/0.0296	0.6250/0.6250/0.6250
MTAD-GAT (ML4ITS)	C/C/C	0.0013/0.0010/0.0013	0.8750/0.8750/0.8750	0.0022/0.0018/0.0022	0.9375/0.9375/0.8750
CATCH	C/C/C	0.0152/0.0399/0.0946	0.3687/0.3317/0.3561	0.0176/0.0348/0.0177	0.8472/0.8750/0.7847
Anomaly Transformer	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–

Table S52: Mac SMD074 explanation outcomes and seed-specific metrics.

Configuration	Status	I_s(37/38/39)	AP_s(37/38/39)	I_t(37/38/39)	AP_t(37/38/39)
DynSphere-E1	C/C/C	0.2835/0.2835/0.2835	0.2943/0.2943/0.2943	0.9845/0.9845/0.9845	0.2500/0.2500/0.2500
PCA reconstruction	C/C/C	0.1159/0.1159/0.1159	0.3640/0.3640/0.3640	0.9246/0.9246/0.9246	0.6250/0.6250/0.6250
KMeans distance	C/C/C	0.0042/0.0041/0.0041	0.5263/0.5263/0.5263	0.8969/0.8972/0.8972	0.6250/0.6250/0.6250
Ridge AR(1)	C/C/C	0.1816/0.1816/0.1816	0.6605/0.6605/0.6605	0.4342/0.4342/0.4342	0.2500/0.2500/0.2500
DynSphere-E4	C/C/C	0.2773/0.2773/0.2773	0.2934/0.2934/0.2934	0.9966/0.9966/0.9966	0.2500/0.2500/0.2500
DynSphere-S1	C/C/C	0.1660/0.1660/0.1660	0.4675/0.4675/0.4675	0.4814/0.4814/0.4814	0.2500/0.2500/0.2500
DynSphere-S4	C/C/C	0.3164/0.2228/0.2228	0.5327/0.5271/0.5271	0.4506/0.4485/0.4485	0.2500/0.2500/0.2500
DAGMM (TAB)	C/C/C	0.1544/0.0464/0.3487	0.3255/0.5804/0.4536	0.0074/0.4971/0.0188	0.4375/0.4375/0.2500
TranAD	C/C/C	0.0017/0.0166/0.0055	0.5881/0.7556/0.5791	0.9929/0.9921/0.9235	0.5000/0.3125/0.5000
OmniAnomaly (TSB-AD)	C/C/C	0.2187/0.2069/0.1901	0.0873/0.0809/0.0788	11.6342/12.7715/13.6745	0.2500/0.2500/0.2500
MTAD-GAT (ML4ITS)	C/C/C	0.0095/0.0093/0.0062	0.5263/0.5620/0.5263	0.4164/0.5034/0.2573	0.3125/0.3542/0.5000
CATCH	C/C/C	0.0359/0.1680/0.0306	0.4126/0.4531/0.4787	0.4935/0.5346/0.6287	0.2500/0.2500/0.2500
Anomaly Transformer	T_F/T_F/T_F	–/–/–	–/–/–	–/–/–	–/–/–

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